MAGIC with Partial Re-Wedged & Wedge Shimming and Electrical Testing

DYNEGY INC.

Equipment Serial #: 270T483

Job Start Date: 03/01/2018

Approved Date: 09/04/2018

Oracle Project ID: A-0034176 | FSP-143192 | C-10301282

Prepared By

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This report may contain confidential and proprietary information subject to a confidentiality agreement.

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1. Summary

1.1 Site Personnel

Name	Category	Role
Carlos Rivero	GE	Senior Customer Contract Manager
Chau Tran	Contractor	FieldCore Generator Specialist
Paul Tang	Contractor	FieldCore GS
Ali Rahimi	Contractor	FieldCore ST Lead Engineer
Art Diaz	Contractor	FieldCore Outage Manager
Jeff Richardson	Contractor	APM DS Superintenden
Jonathon Meacham	Contractor	FieldCore Magic Technician
Joshua Carothers	Customer	MLPP DS Rep.
Terry Parker	Customer	Dynegy Rep.
Mike Robinson	Customer	Dynegy Generator Rep

1.2 Executive Summary

This report documents the generator field in MAGIC Jr inspection (Miniature Air Gap Inspection) performed on unit 290T483 at Dynegy Corp. Moss Landing Power Plant located in Moss Landing, CA from March 01 to Apr 29, 2018. The intent was to provide a condition assessment of generator stator and field with minimal generator disassemble.

GE Power Service provided the following services: Generator Specialist for Technical Direction and LES
Technician operating the MAGIC equipment. Atlantic Plant and Maintenance (APM) supplied manpower during the disassemble, inspection, and reassemble of the unit. The scope of work was
executed on one 10-hour shift per day and seven days per week.

Major work accomplished during this outage included:

- MAGIC visual inspection of stator and field
- MAGIC full stator wedge tightness test
- Visual inspection of stator core and CE & TE end-winding
- Stator DC electrical test: RTD element and 500 VDC insulation resistance measurements, armature winding resistance, and 5000 VDC insulation resistance with polarization index.
- Field electrical test series: winding resistance, AC impedance test, and 500 VDC insulation resistance with polarization index

Emerging work performed in this this outage included:

- ELCID
- Partial generator re-wedging and wedge shimming
- Wedge tightness mapping

ER case submitted during this outage, AKA PAC case, ER-2018315-1562 - On problem with loose and hollow wedges. GE Engineer recommended to repair this loose and hollow wedges problem with a combination of re-wedging and wedge shimming procedures.

Overview:
Based on the results of all field and armature electrical tests, inspection, and wedges repairs, this generator was acceptable for continue service.

All visual inspections and electrical tests were performed in accordance with GEK 103566 revision L

Detailed inspection information, test results and other applicable recommendation are included herein.

A GE publication (GEK 103566 Rev L) for recommended generator inspection and maintenance interval has been included for future reference.

1.3 Work Scope

Gas Turbine	Steam Turbine	Generator Mechanical	Support Systems
N/A	N/A	End Shield(s) Off	N/A

N/A	N/A	Major (rotor out)	N/A
N/A	N/A	Test	N/A
N/A	N/A	Partial Re-wedge and Shimming	N/A

2. Technical

2.1 Recommendations

High Voltage Bushing

High voltage bushings and stand-off insulators should be inspected for cleanliness as well as the presence of any cracks or deformities at any minor or major outage.

Generator Field Assembly

Visually inspect the field at every Major outage or MAGIC inspection IAW GEK 103566. Ref. L

Field Assembly

Perform a standard field test series at each Minor and Major outage IAW GEK 103566 Revision L. The results should be compared to previous test results for indications of winding and insulation degradation over time

Generator Stator

Conduct visual inspection of stator core by utilized MAGIC inspection method at the next minor or major outage

Generator Stator Wedges

Perform wedge tightness assessment on a next outage opportunity

Generator Stator Electrical Testing

Conduct a standard suitability-for-service- DC test at any minor or major outage IAW GEK 103566 Rev. L

Generator End-winding

Conduct visual inspection of CE at the next opportunity outage.

3. Components

GENERATOR - FIELD/ROTOR

FIELD / ROTOR

3.1 GENERATOR FIELD ASSEMBLY

Part Condition: Good

CBM: NA

Part Condition Description

A thorough inspection of the Generator field was performed during this outage using the MAGIC robot high resolution cameras and borescope. Also, the field was inspected upon removed from stator.

The field body balance weights looked properly staked and secure. The pole faces were in good condition with no additional signs of over-heating, arcing or negative sequence currents. There was no evidence of movement or cracking in the field wedges. The coil end turns were in good condition with good ventilation, insulation and coil form.

Recommendation

Visually inspect the field at every Major outage or MAGIC inspection IAW GEK 103566. Ref. L

ST1 GENERATOR SIGNED TEST AGREEMENT.PDF

Hipotential Testing Agreement

Steam Generator **Stator**

Date: 3/10/2018 Generator Serial No. 290T483 Prepared by: Chau Tran

Customer: Dynegy - MLPP Turbine Serial No. N/A

Generator Rated Voltage (E): 18.0 KV

STATOR TESTING (Put an "X" in the box(s) of the test(s) to be performed)

- [X] 1) DC Leakage Test
- [ ] 2) AC Hipotential Test (60 Hz)
- [ ] 3) AC Hipotential Test (0.1 Hz)
- [X] 4) DC Hipotential Test

OVER VOLTAGE PROOF TESTING

GE RECOMMENDED TEST VOLTAGES (1.5E)

	Multiplier	Conversion Factors			
DC Leakage Test:	2.0			36.0	KV DC
AC Hipot Test (60 Hz):	1.5				KV AC
AC Hipot Test (0.1 Hz):	No Longer Offered			N/A	KV AC
DC Hipot Test:	1.5	1.7		45.9	KV DC

- [ ] I **agree** with the use of the above GE Recommended Test Voltages for this unit. _____ (Initials)
- [✓] I **disagree** with the use of the above GE Recommended Test Voltages. I understand that IEEE recommends 1.25E to 1.5E for an Over Voltage Test multiplier. I request the following multipliers: MR (Initials)

CUSTOMER REQUESTED VALUE (Multiplier) FOR DC LEAKAGE TESTING: 1.50

CUSTOMER REQUESTED VALUE (Multiplier) FOR AC/DC HIPOT TESTING: 0.00

CUSTOMER REQUESTED TEST VOLTAGES

	Multiplier	Conversion Factors			
DC Leakage Test:	1.50			27.0	KV DC
AC Hipot Test (60 Hz):					KV AC
AC Hipot Test (0.1 Hz):	No Longer Offerered			NA	KV AC
DC Hipot Test:	0.00	1.7		0.0	KV DC

- [✓] Customer Requested Test Voltage 27.0 KV DC (DC Leakage) MR (Initials)
- [ ] Customer Requested Test Voltage _____ KV AC (60 Hz) _____ (Initials)
- [ ] Customer Requested Test Voltage NA KV AC (0.1 Hz) _____ (Initials)
- [ ] Customer Requested Test Voltage 0.0 KV DC _____ (Initials)

Print Full Name Mike Robinson Title Engr Programs Mgr

Signature Mike Robinson Date 3/10/18

MLPP_Electrical_Test_Consent_270T483.xlsx *GE Energy Services* *Company Proprietary Services*

ST1 GEN SIGNED POST REPAIR TESTING AGREEMENT.PDF

Hipotential Testing Agreement

Steam Generator **Stator**

Date 4/5/2018 Generator Serial No 290T483 Prepared by Chau Tran

Customer Dynegy - MLPP Turbine Serial No N/A

Generator Rated Voltage (E): 18.0 KV

STATOR TESTING (Put an "X" in the box(s) of the test(s) to be performed)

- [X] 1) DC Leakage Test
- [ ] 2) AC Hipotential Test (60 Hz)
- [ ] 3) AC Hipotential Test (0.1 Hz)
- [X] 4) DC Hipotential Test

OVER VOLTAGE PROOF TESTING

GE RECOMMENDED TEST VOLTAGES (1.5E)

	Multiplier	Conversion Factors			
DC Leakage Test:	2.0			36.0	KV DC
AC Hipot Test (60 Hz):	1.5				KV AC
AC Hipot Test (0.1 Hz):	No Longer Offered			N/A	KV AC
DC Hipot Test:	1.5	1.7		45.9	KV DC

- [ ] I **agree** with the use of the above GE Recommended Test Voltages for this unit. _____ (Initials)
- [✓] I **disagree** with the use of the above GE Recommended Test Voltages. I understand that IEEE recommends 1.25E to 1.5E for an Over Voltage Test multiplier. I request the following multipliers: PHS (Initials)

CUSTOMER REQUESTED VALUE (Multiplier) FOR DC LEAKAGE TESTING: 2.00

CUSTOMER REQUESTED VALUE (Multiplier) FOR AC/DC HIPOT TESTING: 1.25

CUSTOMER REQUESTED TEST VOLTAGES

	Multiplier	Conversion Factors			
DC Leakage Test:	2.00			36.0	KV DC
AC Hipot Test (60 Hz):					KV AC
AC Hipot Test (0.1 Hz):	No Longer Offerered			NA	KV AC
DC Hipot Test:	1.25	1.7		38.3	KV DC

- [✓] Customer Requested Test Voltage 36.0 KV DC (DC Leakage) PHS (Initials)
- [ ] Customer Requested Test Voltage _____ KV AC (60 Hz) _____ (Initials)
- [ ] Customer Requested Test Voltage NA KV AC (0.1 Hz) _____ (Initials)
- [✓] Customer Requested Test Voltage 38.3 KV DC PHS (Initials)

Print Full Name PUTRA SUWITO Title PROJECT ENGINEER

Signature _____ Date 4/5/2018

MLPP_Post Repair Electrical_Test_Consent_290T483 GE Energy Services Company Proprietary Services

GEK103566_L.PDF

GE Power

GEK 103566L
Revised, February 2017

Creating an Effective Generator Maintenance Program

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or Inspection. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes the matter should be referred to General Electric Company. These instructions contain proprietary information of General Electric Company, and are furnished to its customer solely to assist that customer in the installation, testing, operation, and/or Inspection of the equipment described. This document shall not be reproduced in whole or in part nor shall its contents be disclosed to any third party without the written approval of General Electric Company.

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Part 1 – Maintenance Outage Planning

1. Background

A thorough maintenance program is recommended to retain generator reliability and avoid major failure expenses. The three important elements of a thorough program are maintenance frequency, visual inspection and testing. It is the intent of this document to provide information on each of these elements, which will assist the owner/operator to establish a thorough and cost effective maintenance program synchronized with the prime mover. Plant operating experience, plant history, previous maintenance or component specific recommendations may make additional scope applicable to an inspection. It is the intent of these recommendations to remove the generator rotor only when a repair requires rotor removal for necessary access. Some generators may have design features that limit robotic inspection, in which case the rotor should be removed. Your local GE service representative can assist in maintenance planning.

During the first several months of operation, the stator winding support system and some of the other generator components experience a break-in period. Therefore, the first maintenance inspection is recommended per Table 2. The most common type of wear that is normally noticed is the dusting in the generator stator end winding. Some amount of light/minor dusting is expected during the initial months of operation as the stator assembly components are subjected to compression, shrinkage, etc. The first maintenance inspection will identify and correct the potentially increased wear. Delaying or omitting the first maintenance inspection can cause increased maintenance outage time, increased maintenance scope, or possibly a forced outage prior to the scheduled maintenance outage.

Subsequent planned outages should be performed in a timely fashion. Experience has shown that regularly scheduled inspection & maintenance outages, as detailed in Table 1, are one of the most important steps in retaining unit reliability and reducing future repair/failure expenses. Borescopic and robotic maintenance outages are recommended per Table 2 and each includes a comprehensive series of electrical tests and thorough visual inspection per Table 3,4 and 5. Each test has its specific advantage and no one test is sufficient to provide complete understanding of the generator health.

Generator monitoring and diagnostics options are also available (Section 8) to trend and monitor generator operational parameters and can provide essential information about generator health and aging. In conjunction with scheduled inspection and maintenance intervals, generator monitoring and diagnostics can help identify normal vs. abnormal/accelerated aging of the generator and reduce future repair/failure expenses. Some examples that may point to the abnormal aging of the generator includes repeated heavy dusting on the stator end windings, early or frequent stator re-wedge, blocked vent ducts in the rotor winding, repeated oil ingress, red dust on stator core, corona activity, etc. In the event that abnormal aging is detected, repair may be necessary and inspection and maintenance practices should be reviewed and tailored to the unit's operating profile.

Generators wear differently in continuous duty application and cyclic duty application. Thermal mechanical fatigue is the dominant life limiter for peaking machines, while insulation breakdown is the dominant life limiter for continuous duty machines. Interactions of these mechanisms are considered in the GE design criteria but are considered second-order effects. If the unit's operation changes due to dispatch needs (continuous duty vs. cyclical duty) inspection & maintenance will be impacted, contact your GE service representative for support. GE bases generator maintenance requirements in Table 1 on independent counts of hours, starts and turning gear hours. Whichever criteria limit is first reached determines the inspection & maintenance interval. These recommendations for inspection fall within the design life expectations and are selected such that components acceptable for continued use at the inspection point will have low risk of failure during the subsequent operating interval.

As a generator continues to age and accumulate operating hours, generator maintenance becomes a critical part of maintaining generator reliability, availability and maintainability. In addition to maintenance, generator monitoring also becomes essential to not only help extend generator life, but also provide critical information on generator stator and rotor health which may be useful for planning future outages.

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2. Maintenance Planning

Advanced planning for maintenance is necessary for generator operators to maintain reliability and availability. The correct implementation of planned maintenance provides direct benefits in the avoidance of forced outages, unscheduled repairs, and downtime.
The basic design and recommended inspection and maintenance for heavy-duty generators are oriented toward:

- Maximum periods of operation between inspections
- On-site, rotor in-situ maintenance
- Minimal disassembly and reassembly of generator components

In addition to maintenance of the basic electrical generator, auxiliaries require periodic maintenances as well. Auxiliaries which should be reviewed for Inspection at the same time as the generator include, but are not limited to:

- Excitation system
- Stator Cooling Water System
- Lubricating Oil and Hydrogen Seal Oil Systems
- Air or Hydrogen Coolers
- Hydrogen Control Cabinet and associated piping
- Generator Control and Protection

Note: Special attention should be paid to TEWAC units with coolers on top. Unmitigated water leaks have led to catastrophic failures.

This document is focused on maintenance planning for the basic electrical generator, which includes the rotor, stator and frame. Alterrex exciters should follow all appropriate recommendations within this GEK including, but not limited to, Inspection and Maintenance Intervals, and Stator and Rotor Test and Inspections. The inspection, maintenance and repair requirements, outlined in the O&M Manual provided to each owner, lend themselves to establishing a pattern of inspections. These inspection patterns will vary from site to site due to numerous factors such as duty cycle and ambient conditions. GE monitors operating experience, inspection results and in-service operating issues of the installed base to the degree that the information is available. Additional information is gathered from units with a connection to a GE monitoring center.

GE uses the data it gathers to analyze and identify potential issues specific to similar units across the fleet with subsequent recommendations to owners on specific matters forwarded by means of Technical Information Letters (TIL) so applicable action can be taken to obtain maximum reliability, availability, and maintainability. The recommendations contained in TILs take precedence over guidelines outlined in this GEK and should be factored into the overall Inspection and maintenance planning program.

A few examples of the primary factors that affect the maintenance planning process are shown as follows and should be integrated into the overall Inspection program:

- O&M manual
- Manufacturer's recommended maintenance program
- Technical Information Letters
- Customer Information Bulletins
- Operational history
- Monitoring trends
- Previous inspections
- Design features
- Duty cycle
- Ambient and environmental conditions
- Risk tolerance
- Reliability/availability needs

Operational events which may impact the generator reliability and availability are listed below. These events should be documented in the plant's operational log. These events can vary in severity and should be considered when planning an outage. The events below are cumulative and may reduce the interval between maintenance or necessitate added scope.

a. Errors in synchronizing
b. Under or over frequency operation
c. Lightning surges or other transients
d. Faults (such as short circuits) on system or adjacent machines
e. Unbalanced load, including single phase operation (Negative Sequence and Motoring events)
f. Overload
g. Over voltage or under voltage
h. Loss of rotor excitation
i. Loss of cooling water without load run-back
j. Asynchronous operation
k. Overspeed

Each of these events will have a different impact on the necessary planning for maintenance and inspection. Depending on the circumstances, some of these events might dictate that the machine be shut down for immediate inspection, whereas other events may allow a reasonable wait for the next scheduled inspection. Contact your local GE service representative for recommendations associated with these events.

The maintenance program should reflect the level of acceptable risk for the unit. This will vary from unit to unit and plant to plant, and will change over time as the importance of the unit to the power system changes. In addition, new technologies are constantly being developed to improve unit reliability, performance, monitoring and inspection equipment, to provide a cost-effective means for maintaining the generator. The in-situ robotic inspection is one example of technology developed by GE to provide a more cost-effective means of performing a generator inspection. Other examples include advanced continuous monitors such as stator partial discharge, flux probes, and collector health. The owner/operator should be aware of these developments and modify their maintenance program accordingly. While this should be a continuous process, the "Maintenance outage planning review" is an appropriate checkpoint.

Planned maintenance outages are usually scheduled well in advance of the actual outage date, and preparation for the outage should begin early. Materials that might be needed for the maintenance should be ordered in advance so that they will be available at the start of the outage to avoid the risk of a costly delays. The local GE service representative can assist in maintenance planning including incorporating any appropriate new upgrades or inspection techniques. Identification of strategic spares based on unit, site or fleet performance is also possible.

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3. Inspection and Maintenance Frequency

There are many components which require routine maintenance or inspections between scheduled outages. The operator will find these recommendations in various supplier's equipment manuals and should include additional maintenance tasks as operating experience indicates. Results of routine maintenance should be retained in well-organized files readily available for reference. These routine maintenance records coupled with the information from the monitored operating data are a good indicator of pending service or operating problems that should be addressed at the next scheduled outage.

The frequency and level of inspection and maintenance can be dependent upon service duty, system demands, age of the unit and risk tolerance among many other individual plant considerations. These factors were discussed in Section 2 Maintenance Planning. The owner must balance performance, reliability and cost when planning outages and setting maintenance and inspection intervals. Many of the factors related to the timing of inspections are determinable by the owner/operator.

In the absence of operating experience and resulting condition assessments, Table 2 lists the GE recommended inspection intervals for generators. Generally speaking, generators can be inspected commensurate with the associated turbine outage schedule. These initial intervals should be reviewed and adjusted as operating and inspection and maintenance experiences are accumulated. Reductions in the recommended intervals could result from abnormal operational events described previously or due to unfavorable operating experience. Increases in the recommended intervals may also be considered where operating experience has been favorable. Historical operation and machine conditions can be used to optimize maintenance programs for a specific site/machine. GE can assist operators in determining the appropriate maintenance and inspection intervals for their operating profile.

During turning gear operation relative motion occurs in the rotor winding components potentially causing fretting wear. It is advised to the operator to minimize turning gear operation to the extent possible without violating limits on the other rotor train components. Some turning gear systems rotate the rotor at a high enough speed that fretting wear does not occur, contact GE to assist in assigning appropriate maintenance and inspection intervals or upgrades. Otherwise, reference turning gear inspection and maintenance intervals listed in table 2.

In addition to offline inspections and maintenance, there are online tests that can be performed to provide insight to the unit health and scope of upcoming outages. One such test is the use of flux probes to monitor the relative magnetic flux and determine if the field winding has experience shorted turns. If equipped with flux probes, at minimum a flux probe test should be performed in the planning stages for an inspection outage to help define the outage scope. Additionally, it is recommended to perform such tests on a more frequent basis such as yearly to allow trending of any degradation that might occur over the life of the field winding.

It is recognized that some power generation plants may run in an operational regime which could produce extended turbine maintenance outage intervals in the 6 to 10-year range. Regardless of plant operating regime, GE can help the owner define a customized cost effective maintenance plan. If the generator has 6-10 years between maintenances with significant time on turning gear, then there is a potential impact on rotor winding life and maintenance interval should be discussed with the local GE service representative.

During a **First Inspection**, a Borescopic Inspection is recommended to be performed, typically using access via a port, manway or by removing a cooler. The upper end shields or end plates do not require removal to permit inspection of the end winding area and the rotor remains in place. For those generators where access can not be gained to insert a borescope; removal of one or more end-shields may be required to complete the visual inspection. Additionally, a series of low voltage tests should be conducted in accordance with Tables 3, 4, and 5.

A ***Borescope Inspection*** will typically access the internals via manway or by removing a port or cooler. The upper end shields or end plates may not need to be removed to permit inspection of the end winding area and the rotor remains in place. For those generators where access cannot be gained for insertion of a borescope; removal of one or more end-shields may be required to complete the visual inspection. Additionally, a series of high and low voltage tests should be conducted in accordance with Tables 3, 4, and 5.

A ***Robotic Inspection*** is capable of performing a thorough inspection of the stator and rotor. Robotic inspection of the rotor and stator may require the removal of a few entrance gap baffles to allow access to the air gap with the rotor in place. The inspection includes a series of electrical tests and a visual inspection. Note that robotic inspection may not eliminate the need to remove the rotor if an issue is discovered that requires rotor removal to correct, such as loose stator core wedges or significant end winding maintenance or repairs. Inspection intervals are based on the age of the stator. For those units not accessible by a robot, more frequent Borescope Inspections, pulling the rotor, use of monitoring or some combination of these options should be used. In such a case, contact your local GE service representative for assistance in optimization of inspection and maintenance intervals and scope.

While following maintenance and inspection intervals define within this GEK, conditions may dictate that a future outage will require rotor removal for the maintenance or repair of the rotor or areas of the stator which are inaccessible with the rotor installed. The need to remove the rotor may be predicated by a single event or long term trends, examples of such are listed in Table 1. For support in planning the scope of future outages, contact your local GE service representative for assistance.

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Table 1 Rotor Removal

Events or Trends Predicating Rotor Removal	Required Maintenance or Repair Activities
Abnormal operating events as listed in Section 2: Maintenance Planning	Rotor tooth and wedge inspection and repair to remove hardened material
Rotor vibration trends coupled with flux probe shorted turns or blocked ventilation passages	Rings off cleaning up to field rewind
Stator wedge looseness trends	Partial or full stator re-wedge
Air gap foreign object damage	Core or wedge repair
Stator winding insulation degradation indicated through DC leakage or Partial Discharge trending	Full stator rewind

During the course of normal inspection and maintenance intervals, there is the potential for inspection findings to require the removal of the rotor for maintenance and repair activities. While uncommon, accelerated aging of generator components may occur requiring the removal of the rotor to perform repair activities. Foreign object damage, oil ingestion events, and contamination from external environment for air cooled units can also result in inspection findings requiring more intrusive repairs if preventative maintenance per O&M manuals is not strictly followed. GE's experience has shown with proper trending and planning; it is unlikely that a planned Borescope or Robotic Inspection will require an unplanned rotor removal.

Tables 3 through 5 display recommended tests and inspections.

In general, it is recommended that the initial Inspection and Maintenance Interval, as listed in Table 2, be a Robotic inspection with subsequent intervals alternating between Borescopic and Robotic Inspections. Reference Figures 1 through 3 for CC and ST plant generator inspection and maintenance interval examples. These recommendations are based on fleet operating experience and are intended to optimize the outage duration and intrusiveness after the initial break in period of the generator components. After 160,000 hours of operation or time on turning gear, 25 calendar years or 5,000 starts it is recommended to only perform robotic inspections. As a generator ages beyond 25 years it may require a shift to a more frequent maintenance interval. Figure 2 provides an example where the borescopic inspection has been aligned with the Hot Gas Path inspection via utilizing a full robotic inspection in place of the first inspection with a borescope.

During a stator rewind, all static component should be inspected and replaced or repaired as necessary resulting in a reset of the components age and inspection regimen. As the stator core is exposed during a rewind, a detailed inspection of the core is recommended including at minimum a visual inspection, EL-CID and core tightness testing. However, this is also an opportune time to perform a core ring test and retighten of core compression features if applicable. Because the rotor, stator and other components age at different rates and are typically rebuilt/rewound at intervals different from that of the stator, more frequent maintenance intervals may be appropriate after a stator rewind. Any questions about maintenance intervals should be discussed with the local GE service representative

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Table 2 Recommended Inspection and Maintenance Intervals[1,2,3,4]

	First Inspection	Inspection and Maintenance Intervals
All Units not listed below	8,000 Turbine Hours; 250 Starts; 4,000 Turning Gear Hours	24,000 to 32,000 Turbine Hours; 900 to 1,250 Starts; 30,000 to 32,000 Turning Gear Hours
H55 (330H) Generators H73 (390H) Generators H75 (450H) Generators LC generators w/phosphorous brazed stator bars. Generators older than 25 years that have not been rewound with updated materials.	8,000 Turbine Hours; 250 Starts; 4,000 Turning Gear Hours	20,000 to 24,000 Turbine Hours; 750 to 900 Starts; 22,000 to 24,000 Turning Gear Hours
In the absence of operating experience and resulting condition assessments, Table 2 lists the recommended inspection intervals for turbo-generators. Generators can be inspected commensurate with the associated turbine inspection outage. These initial intervals should be reviewed and adjusted as operating and inspection experiences are accumulated. LC-Liquid Cooled		

Note:

[1] Inspection and maintenance intervals reflect operation and inspection in accordance with manufacturer specifications. They represent initial recommended intervals in the absence of operating and condition experience.

[2] Factored turbine hours, factored turbine starts are calculated in accordance with turbine manufacturer specifications.

[3] Inspection and maintenance intervals are determined by whichever occurs first, whether it is actual turbine hours, factored turbine hours, actual turbine starts, factored turbine starts, or turning gear hours.

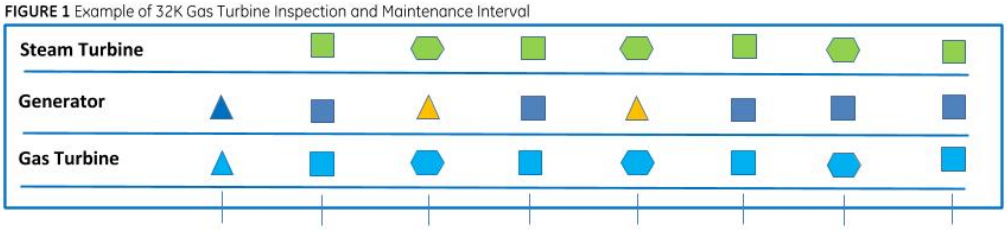
[4] Some turbine manufacturers do not specify factored turbine hours or factored turbine starts. Actual turbine hours and actual turbine starts should be utilized in those specific circumstances.

Example Inspection and Maintenance Interval Charts can be seen in Figures 1 through 3 below.

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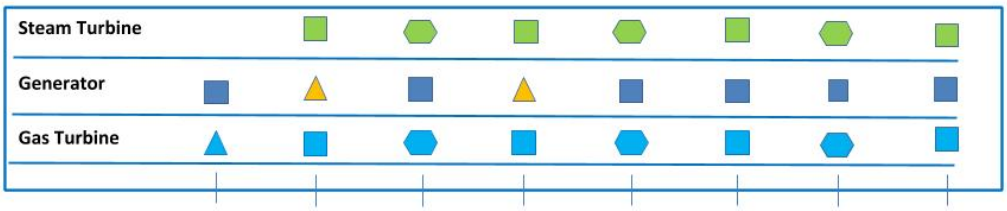
FIGURE 1 Example of 32K Gas Turbine Inspection and Maintenance Interval



Factored or Actual Turbine hours (hrs.)	8K	32k	64k	96k	128k	160k	184k	208k
Gen Insp Type	1st	Robotic	Borescope	Robotic	Borescope	Robotic	Robotic	Robotic
Expected Interval for 6K hrs per year units	≤18 months	5 yrs.	10 yrs.	16 yrs.	21 yrs.	26 yrs.	30 yrs.	34 yrs.

For this specific example, it is assumed that the turbine generator Inspection and maintenance interval is determined by factored hours or actual hours, typical of a unit that is operating over 6000 hours per year.

FIGURE 2 Example of Alternate 32K Gas Turbine Inspection and Maintenance Interval



Factored or Actual Turbine hours (hrs.)	8K	32k	64k	96k	128k	160k	184k	208k
Gen Insp Type	1st	Robotic	Borescope	Robotic	Borescope	Robotic	Robotic	Robotic
Expected Interval for 6K hrs per year units	≤18 months	5 yrs.	10 yrs.	16 yrs.	21 yrs.	26 yrs.	30 yrs.	34 yrs.

For this specific example, it is assumed that the turbine generator Inspection and maintenance interval is determined by factored hours or actual hours, typical of a unit that is operating over 6000 hours per year. Borescopic inspections are aligned with Hot Gas Path inspection.

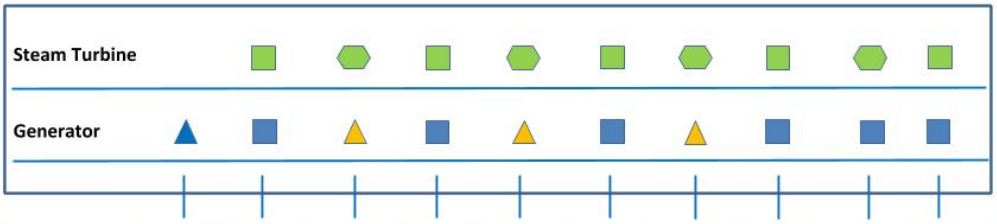
Generator	**1st Inspection**	**Borescope Inspection**	**Robotic Inspection**
Gast Turbine	**Borescope Inspection**	**Hot Gas Path**	**Major Inspection**
Steam Turbine		**Minor Inspection**	**Major Inspection**

Consult the appropriate maintenance and inspection manual for interval guidance. This example is meant to demonstrate how generator inspection and maintenance intervals would be synchronized for a gas turbine with the referenced intervals.

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FIGURE 3 Example of Fossil Coal Steam Turbine Inspection and Maintenance Intervals (24K EOH Inspection Interval)



Effective Operating Hours (EOH)	8K	24k	48k	72k	96k	120k	144k	168k	192k	216k
Gen Insp Type	1st	Robotic	Borescope	Robotic	Borescope	Robotic	Borescope	Robotic	Robotic	Robotic
Expected Interval for 8K EOH per year units	≤18 months	3 yrs.	6 yrs.	9 yrs.	12 yrs.	15 yrs.	18 yrs.	21 yrs.	24 yrs.	27 yrs.

EOH – Effective Operating Hours. Please consult the guidance of the steam turbine OEM.

Generator	1st Inspection	Borescope Inspection	Robotic Inspection
Steam Turbine		Minor Inspection	Major Inspection

Consult the appropriate maintenance and inspection manual for interval guidance. This example is meant to demonstrate how generator inspections would be synchronized for a steam turbine with the referenced intervals.

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Table 3 Recommended Stator Tests and Inspections

Stator and Core			AIR	HYD	LCSW	Inspection		
Test	Component	Inspection Objectives and Assessment				First Inspection	Borescope	Robotic
RTD Element Resistance	Gas and Winding RTD's	Checks for calibration and poor connections	X	X	X			X
RTD Ground Insulation	Stator Winding RTD's	Insulation condition of RTD	X	X	X			O
Winding Copper Resistance	Stator Winding	Checks for poor connections and breaks	X	X	X	X	X	X
Insulation Resistance (aka Megger™)	Collector End (CE) Bearing	Contamination and/or deterioration of insulation	X	X	X			X
Insulation Resistance (aka Megger™)	Hydrogen Seal Casing	Contamination and/or deterioration of insulation		X	X			X
Insulation Resistance (aka Megger™) and Polarization Index (PI)	Stator Winding	Contamination and/or deterioration of insulation	X	X	X[6]	O	X	X
DC Leakage Current	Stator Winding	Contamination and/or deterioration of insulation	X	X	X[6]	O[7]	X	X
DC Hipot	Stator Winding	Ground wall insulation integrity	X	X	X[6]	O	X	X
Wedge Tightness Map[1]	Stator Wedges	Detect Wedge tightness deterioration	X	X	X			X
Visual Inspection	All Visible Components within the End Winding Compartment	Cleanliness, foreign material, loose or displaced parts, damage, wear, dusting, corrosion, or deterioration	X	X	X	X	X	X
Endwinding Tightness Check[8]	Endwinding support rods	Torque measurement	X	X	X			X
Visual Inspection[3]	High Voltage Bushings	Surface condition and wear, cracks, burning, oil contamination, deterioration	X	X	X	X		X
Visual Inspection[10]	Bearings and Seals	Suitability for Service	X	X	X			X
Visual Inspection	Coolers	Suitability for Service	X	X	X			X
Partial Discharge	Stator Winding Insulation	Localized deterioration	X	X	X	O[7]	O-R[5]	O-R[5]
Water Flow Verification[4]	Water Cooled Stator Winding	Restrictions in hydraulic circuit			X			O
Low Flux (EL CID)	Stator Core Insulation	Weak or damaged core enamel	X	X	X			O
Differential pressure measurement [9]	Water cooled stator winding	Restrictions in hydraulic circuit			X		X	X

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Table 3 Recommended Stator Tests and Inspections

Stator and Core			AIR	HYD	LCSW	Inspection		
Test	Component	Inspection Objectives and Assessment				First Inspection	Borescope	Robotic
Dynamic Frequency response	Stator End Winding	Potentially damaging resonance	X	X	X			O
Core Ring Test	Stator Core Insulation	Weak or damaged core enamel	X	X	X			O
Tan Delta	Stator Winding	Stator insulation degradation	X	X	X			O
AC Hipot[2]	Stator Winding	Ground wall insulation integrity	X	X	X			O

Air – Air Cooled Generator
HYD – Hydrogen Cooled Generator
LCSW – Hydrogen Cooled Generator with Liquid Cooled Stator Winding

X - Pertains only to type of unit selected
R - These tests are performed while the unit is running.
O – Optional

[1] This test does not apply to units with asphalt stator windings or units with the "camelback" wedge system. Some model generators may require wedge testing at more frequent intervals, consult O&M manual for guidance.

[2] This test is performed as standard during new unit construction or stator rewind. DC Hipot may be used at customer's discretion during stator rewinds.

[3] Consider replacement of high voltage bushings around 30 years of age and minimize removal/reinstallation to prevent wear that could increase hydrogen leakage on hydrogen cooled generators.

[4] This test is recommended during robotic inspections while the end-shields are removed. Variation in header flow can be an early indicator of the build-up of copper oxides and the resulting reduction of cooling flow to the stator bar. Units for which testing is strongly encouraged are those that have seen a departure from commissioning temperatures or a departure from nominal stator cooling water flow/pressure drop across stator units with only partial temperature monitoring of stator bars, units that have had chemistry excursions (elevated copper concentration, conductivity or variation in oxygen concentration) or have had leakages (water-to-hydrogen leakage or hydrogen-to-water leakage).

[5] For hydrogen cooled generators Off-Line Partial Discharge Testing with the unit open to air may provide additional diagnostic information.

[6] Requires draining and drying of Liquid Cooled Stator bars prior to testing.

[7] DC Leakage and Partial Discharge testing are optional but highly recommended for setting baselines for trending.

[8] This inspection applies only to generators with end winding support systems that can be retightened, this inspection should be performed during every inspection except for the First Inspection. Consult the O&M manual for applicability.

[9] Delta P measurement between water inlet and outlet on units equipped with stator bars using stainless steel hollow conductor technology. A Delta P measurement can also be performed on generators with hollow conductors made from copper if the flow through single bars is not measured.

[10] An equivalent inspection via digital trending and analysis may be suitable, contact your local GE Service Representative.

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Table 4 Recommended Water Cooled Stator Leak Tests for High Oxygen Systems†‡

Stator		Clip-Strand Braze: Phos Containing				Global Epoxy Injection or Clip-Strand Braze: Phos-Free			
H2 Leak Monitoring Method		Orig SLMS, Bagged Vent, or No Monitoring		SLMS HP Installed		Orig SLMS, Bagged Vent, or No Monitoring		SLMS HP Installed	
Test	Inspection Objectives and Assessment	Borescope	Robotic	Borescope	Robotic	Borescope	Robotic	Borescope	Robotic
Vacuum Decay	Checks the hydraulic integrity of the entire winding	X	X			X	X		
Pressure Decay	Checks the hydraulic integrity of the entire winding	X	X			X	X		
Helium Tracer Gas	Detects minute leaks in the hydraulic circuit		X		X		X		
Capacitance Mapping	Wet ground wall bar insulation		O		X		X		
WIM – Wet Insulation Measurement	Wet ground wall bar insulation		X		O		O		

† These tests are required to detect hydrogen leaks in water cooled stator windings only and should be performed in conjunction with the recommended tests in Table 3
‡ Not required for stator bars with stainless steel hollow conductors

Low-Oxygen Stator Cooling Water System

The same tests found in Table 4 can be conducted on Low-Oxygen systems. Where the operations manual recommends maintaining the Stator Cooling Water System (SCWS) in operation then the tests found in Table 4 can not be conducted. Stator electrical testing requires the SCWS to be shutdown and drained. In the event the SCWS is not shut down, conduct a visual inspection for water drops as evidence of hydrogen leaks. If hydrogen leaks are suspected, it is recommended to shut down the SCWS, fully drain and dry the system and conduct the inspections in table 4.

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Table 5 Recommended Rotor Test

Rotor			AIR	HYD	LCSW	Inspection		
Test	Component	Inspection Objectives and Assessment				1st Inspection	Borescope	Robotic
Winding Copper Resistance	Rotor Winding	Checks for poor connections and breaks	X	X	X	O	X	X
Insulation Resistance (aka Megger™)	Rotor Winding	Contamination and/or deterioration of insulation	X	X	X	O	X	X
Shorted Turn Test[1]	Rotor Inter-Turn Insulation	Turn Shorts	X	X	X		X	X
Recurrent Surge Oscillography (RSO) & AC Impedance[2]	Rotor Inter-Turn Insulation	Turn Shorts	X	X	X		O	O
Incremental Rotational Winding Copper Resistance	Main lead, Gooseneck, and Bore Copper	Looseness of main lead and orientation dependent connectivity issues	X	X	X		O	O
Visual Inspection	All Visible Components within the End Winding Compartment	Cleanliness, foreign material, loose or displaced parts, damage, wear, corrosion, deterioration	X	X	X	X	X	X
Visual Inspection	Collector System	Surface condition and wear, cracks, burning, oil contamination, deterioration	X	X	X	X	X	X
Visual Inspection	Journals	Check for suitability of service	X	X	X			X
Optional Tests								
Over Potential/Hipot	Rotor Winding Insulation	Ground wall insulation integrity	X	X	X			O
Bore Pressure Test	Bore / Terminal Seals	Sealing capability of the seals		X	X			O

Air – Air Cooled Generator
HYD – Hydrogen Cooled Generator
LCSW – Hydrogen Cooled Generator with Liquid Cooled Stator Winding

X - Pertains only to type of unit selected
R - These tests are performed while the unit is running.
O - Optional

[1&2] The preferred method for shorted turn testing is the flux probe. Another method is the use of RSO and AC Impedance testing when flux probes are not present. Flux probe testing is performed with the unit online.

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4. Testing

Generator electrical testing is focused on the insulation systems. There are other tests used to monitor for degradation in other components. A list of the typical tests recommended and the test purpose is shown in Tables 3, 4, and 5. Historical records of test results should be maintained and compared to the new test results. Changes between outage test results may point to needed repairs that may not be evident from the absolute test values themselves.

Many of the tests require special equipment that has been properly maintained and calibrated to assure accurate results.

5. Visual Inspection

A visual inspection, performed at every inspection by an experienced individual, can uncover unit conditions previously undetected or validate monitoring findings. In some instances, electrical testing will only detect issues when the stator and rotor insulation systems have been significantly compromised. For example, there is no definitive test for contamination, rust or oil, and water leaks, and yet the presence of any of these could adversely affect reliability and operation. A typical visual inspection should include those items listed in Tables 3 and 5.

As a critical component of the overall maintenance program, visual inspection is limited to areas that can be accessed for view either directly or with mirrors, borescopes, cameras, etc., and therefore must be combined with other tests to give a complete picture of generator condition.

6. Rotor In-Situ Inspection via Robotics

GE's robots provide a thorough visual inspection of the stator core and windings and rotor body using the robot's on-board high-resolution video cameras. Additionally, robots can provide quantitative wedge tightness and Electromagnetic Core Imperfection Detection (ELCID) assessments. Coupled with the Remote Access Camera (RAC), robots can provide a comprehensive stator and rotor inspection. Robotic inspections are performed to cover the tasks that otherwise require the removal of the rotor.

GE robots can currently enter generators with an airgap of 0.36 inches / 9mm or larger. Airgap alone is not sufficient to determine testability of a generator as some features may make rotor removal the preferred choice. Examples include baffles or cooling fan configuration. Check with your local GE service representative if you have questions.

7. Rotor Life Management

Modern generator rotor forging and retaining rings are typically designed for normal transient events including 10,000 start/stop cycles. Earlier generators may have had different start/stop design criteria, particularly large fossil & nuclear units. For older (25+ years) generators that have changed operating profile (base load to cyclic / peaking duty), GE Product Service should be contacted for an appropriate inspection plan. However, with proper inspections and maintenance, generators may last much longer. Common issues with aged rotor forgings are typically related to the number of start/stop cycles. Several Technical Information Letters (TILs) have been written on the subject. Particularly when performing rotor rewinds, those TILs should be reviewed and the inspections performed. These inspections are mostly focused on cracks in the wedge loading face, in the slot and in the bore if the rotor forging has one.

Typically, the insulation system of a field will deteriorate over time and service such that a rewind is required prior to reaching the design life of structural components, at which point a full inspection can and should be performed. Nevertheless, if a unit reaches the design life (10,000 start/stop cycles for modern units) and has not yet had structural components inspected, GE Product Service should be contacted for a unit specific evaluation. Unless otherwise defined following unit specific inspection, the re-inspect interval will be set at half the initial inspection recommendation.

Non-Destructive Testing at a Rotor Rewind

Magnetic particle testing of the external surfaces of the rotor should be performed at outages that expose the bottom of the rotor slot. Magnetic fields in both the axial and circumferential directions should be used. All structural components removed as part of the rewind (wedges, retaining rings, centering rings, radial fans, axial fan hubs, collector rings, coupling rings, etc.) should also be inspected.

Bored Rotors

For units manufactured prior to 1985, applicable TILS and unit specific re-inspect recommendations describe the rotor bore inspection cycle and scope.

For rotors that were manufactured after 1985 with a bore that extends under the full length of the body, it is recommended to perform ultrasonic volumetric testing and a magnetic particle inspection of the bore within 10,000 stop start cycles. Recommendations for subsequent re-inspections will be made based on the test results and associated structural integrity, with a 20 year and 2,000 starts interval being the maximum. If the rotor has not been inspected within the last 20 years or the rotor has operated higher than the re-inspect interval, then the rotor should be inspected at the earliest opportunity. Please note-this does not apply to the collector bore.

Unbored Rotors

For rotors without a bore under the body it is recommended to perform ultrasonic volumetric testing from the periphery, inspecting the core of the shaft under the body within 10,000 start-stop cycles. This inspection can be completed on a fully assembled rotor, nevertheless including the inspection at a rewind may be the most convenient time to do so.

Magnetic particle testing of the external surfaces of the rotor should be performed at outages that expose the bottom of the rotor slot. Magnetic fields in both the axial and circumferential directions should be used. Note that the axial direction may not be possible unless copper windings are removed

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Table 6: Rotor Volumetric Inspections

Generator Rotor Characteristics	Outage Type	Ultrasonic Volumetric	Mag Particle of all shaft diameters and slots
In-service and has a main body bore	Inspection	Notes 1 and 2	Note 3
In-service more than 20 years with no main body bore	Inspection	Note 1	Note 3
In-service and has a main body bore	Rewind	Notes 1 and 2	Note 4
In-service more than 20 years with no main body bore	Rewind	Note 1	Note 4

Note:

[1] Ultrasonic volumetric testing includes boresonic tests (bored rotors only), axial UT and periphery UT tests.

[2] Magnetic particle testing of the rotor bore surface (bored rotor only) should be performed concurrent with ultrasonic volumetric testing.

[3] The turbine end of the generator rotor from the coupling face to the fan ring should also receive a magnetic particle inspection concurrent with ultrasonic volumetric testing. Caution should be exercised during magnetic particle testing to prevent foreign particles from entering the rotor windings.

[4] During full rewinds, perform mag particle testing of all accessible surfaces including axial and circumferential magnetization

Collector Ring Removal/ Replacement

Collector rings may be removed / replaced separately from a rotor rewind. Therefore, during collector ring removal the exposed collector shaft should be NDT inspected. All structural components removed as part of the work scope (collector rings [if reused], couplings, fan hubs, etc.) should be inspected.

Retaining Rings

Current GE generators are equipped 18Mn-18Cr retaining rings. Customers with generators that have 18Mn-5Cr retaining rings still in-service should consider replacing them with 18Mn-18Cr retaining rings at the next scheduled outage. If replacement is not viable in the near future, the risk of stress corrosion cracking on 18Mn-5Cr retaining rings can be mitigated by performing visual inspections during every minor inspection and non-destructive testing during every major inspection. In general, retaining rings in air-cooled generators are at a medium to high risk of developing indications due to stress corrosion cracking, in comparison to retaining rings in hydrogen-cooled generators, due to the moisture and humidity found in the air.

For 18Mn-18Cr retaining rings, GE recommends that retaining ring inner diameter (ID) be inspected using fluorescent penetrant inspection either during rewind or any available opportunity when retaining rings are disassembled from the unit. At the time of writing this document, GE has not encountered any cracking of 18Mn-18Cr rings due to stress corrosion cracking under normal operating modes.

Any abnormal operating events such as those listed in section 2 may require additional review and additional non-destructive inspections at intervals other than specified above. Similarly, presence of halides (fluorides, chlorides, bromides, iodides and astatides) would require additional review and additional non-destructive inspections at intervals other than specified above.

It is recommended that the 18Mn-18Cr retaining rings have Non-Destructive Testing before accumulating 30 years of service or 10,000 start/stops or earlier, if advised by the manufacturer's recommended maintenance program, and every 10 years or every 2,000 starts thereafter. Such testing can be conducted with the rings on or off. Rings on would typically consist of Eddy Current inspection of the outer diameter and Ultrasonic Testing from the outer diameter to examine the inner diameter.

For rotor rewinds, GE recommends to assess the remaining life for re-use; judgement based on accumulated start-stop cycles and findings of the non-destructive testing.

Turbine End Shaft

The turbine end of the generator rotor from the coupling face to the fan ring should also receive a magnetic particle inspection concurrent with bore/yoke ultrasonic inspection. Caution should be exercised during magnetic particle testing to prevent foreign particles from entering the rotor windings.

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Part 2 – Other Maintenance Considerations

8. Other Inspection Considerations – Part 2

Part 1 includes general guidance and recommendations for generator inspections and maintenance. As technology is continually evolving, additional methods that aid the owner/operator in maintaining generator reliability at lower lifecycle costs become available that could provide an opportunity to tailor an inspection program specific to a particular generator Non-invasive inspection, monitoring and diagnostic technologies are a few recent examples.

Monitoring & Diagnostics

Between periods of planned offline Inspection, there are many parameters an operator can monitor to detect changes in equipment condition. Analysis of this operating data can be used to understand the overall condition of the unit and enables prediction of possible equipment failures in advance.

In conjunction with monitoring for abnormal events as listed in Part 1, Section 2 and trending prior testing and inspection history, generator operational parameters can be monitored to trend long term unit health and capability. Periodically, the generator loading should be compared to the generator capability curves to trend changes in capability with cooling temperature variation. These capability trends including resulting stator and field temperatures should be tracked for shifts not associated with operational mode changes. In the event that an abnormal shift occurs which is not associated with intentional operational changes, consult local GE service representative to determine if any actions are necessary. Other operational parameters such as alarm and trip history as well as rotor vibration and bearing metal temperature should be trended long term can provide insight to generator health.

Today's GE Generator Health Monitor enables the monitoring and trending of some key aging and wear mechanisms allowing the customer to have the time to plan inspection timing and scope while managing risk. Some monitoring options include but are not limited to:

Table 7. Monitoring Systems

Monitor	Trend/Detect
Partial Discharge	Stator aging and wear
Flux Probe Monitoring	Number and location of turn-to-turn shorts
Collector Monitor	Trend "sparking events" to prevent flashover
Shaft Voltage Monitor	Deteriorating bearing insulation, shaft rubbing or various causes of asymmetries within the generator, turn-to-turn shorts (not as accurate as flux probe)
End Winding Vibration Monitor	Detect changes to the core end-winding
Liquid Cooled Stator Temperatures	Provide early warning of plugged strands or bars in order to prevent overheating.
Stator Leakage Monitoring System for Liquid Cooled Stators	Detect hydrogen leakage into the Stator Cooling Water System

Should further information be desired or should problems arise which are not covered sufficiently for the owner/operators purposes the matter should be referred to the local GE service representative.

Chemical cleaning

Chemical cleaning is a means of improving stator cooling when flow restrictions have developed in the stator bars over time. Chemical cleaning applies only for stator bars that contain copper hollow conductors. Units with stainless steel hollow-conductor technology do not require chemical cleaning.

Two chemistry control regimes can be used effectively to reduce corrosion in the Stator Cooling Water System (SCWS): high-oxygen and low-oxygen. A low-oxygen system reduces corrosion by limiting the availability of the oxygen that is required for corrosive chemical reactions. Low-oxygen systems may operate with a neutral or alkaline pH. High-oxygen systems manage corrosion through formation and maintenance of a thin, protective oxide layer on copper surfaces. Under both treatment regimes, control of SCWS chemistry and flow is required for system stability, to reduce the rate of copper dissolution, and to minimize particulate deposition. Despite these measures, which may reduce the rate, even a well-controlled SCWS may gradually experience copper dissolution and oxide deposition through normal aging mechanisms. The SCWS has a resin bed and filter to remove ions and particulates, but some can settle out in the cooling strands upstream of the resin bed. Eventually, this phenomenon can lead to a reduction in flow and a temperature increase in the stator. If left unchecked, hollow strands can clog completely, and if enough strands clog, whole stator bars can lose their effective cooling capability—potentially leading to failure of the stator due to overheating

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The SCWS and stator temperatures should be monitored for loss of cooling efficiency. Once detected, cooling losses should be closely monitored and either an on-line or off-line chemical cleaning of the SCWS considered. Several units have experienced slow cooling loss for an extended time, followed by an unexpected, rapid cooling loss caused in part by a feedback loop in which reduced flow caused by oxide deposition raises temperatures, and high temperatures increase the rate of oxide formation. A proactive approach is advised. While generators can operate at reduced load to manage temperatures while arranging for chemical cleaning, such a cleaning relies on the ability to circulate the cleaning chemicals. Once a strand is completely plugged, cleaning chemicals cannot flow through it and it is unlikely that cooling flow can be restored solely by chemical means. In such cases, mechanical cleaning is an option, though experience with mechanical cleaning has not always been positive. The best approach is a proactive approach, where cleaning is conducted early so that the cleaning chemicals can flow through all cooling channels. Key elements for early recognition include monitoring and proper maintenance of the SCWS during outages.

High-Oxygen SCWS

The strategy in a high-oxygen system is the establishment of a protective cupric oxide layer, which forms on copper surfaces under conditions of air-saturated water. When a cupric oxide forms on a copper surface, it forms a boundary that prevents further corrosive chemical reactions that might dissolve copper under the oxide. High dissolved oxygen concentration in the water is maintained either by active aeration, or regular feed and bleed with aerated make-up water. Low dissolved oxygen, acidic pH, and/or improper lay-up during an outage can all destabilize the protective oxide layer. It is thus advised to maintain the SCWS chemistry as stable as possible. A common cause of chemical instability is insufficient aeration, which reduces the oxygen concentration below the level that promotes formation of cupric oxide. Leakage of hydrogen into the SCWS can destabilize the protective oxide both by displacing some of the dissolved oxygen. Excessive carbon dioxide, chloride or other impurities in the water can also reduce pH and damage the cupric oxide.

Low-Oxygen SCWS

The strategy in a low-oxygen system is minimization of oxygen ingress. With low levels of dissolved oxygen, the stable oxide that forms on copper surfaces is cuprous oxide. Unlike the cupric oxide that forms in a high-oxygen system, cuprous oxide formation typically does not cease once a layer has formed. Any oxygen that leaks into a low-oxygen system will react to create more cuprous oxide, so it is essential to maintain stable SCWS chemistry with the lowest possible oxygen level. Proper operation of a low-oxygen system requires preventing the introduction of aerated water, and holding a stable pH.

Common root causes for a departure from normal conditions include elevated consumption of aerated make-up water and improper lay-up or system restoration during an outage. Other sources of oxygen can include direct air leakage into the system, and leakage of hydrogen that has been contaminated by air.

With the goal of early recognition of potential problems, Table 8 lists a selection of SCWS parameters that should be monitored, and a recommended monitoring frequency. Baseline values and an operating history should be recorded for most of these parameters so that adverse trends can be identified in enough time to avoid or mitigate failure.

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Table 8. Detection of flow restriction: recommended Monitoring for High Oxygen Systems (HOS) and low Oxygen Systems (LOS), neutral water chemistry.

Stator cooling water system					Action		
Parameter	Frequency		Typical value[2]		Troubleshooting[3]		Proactive Chem cleaning
	On-line	Interval[1]	LOS	HOS	LOS	HOS	Trend within months
Conductivity	X		0.55 – 0.3 µS/cm		≥0.5 µS/cm		
Oxygen	X		≤ 20 ppb	≥ 2 ppm	> 10 above norm	< 2 ppm	
pH	X		6.5 – 7.5		< 6.5		
Copper		X	5 ppb	5-10 ppb			
Stator bar temperatures	X						T_{max} - T_{avg} > 6 °C[4]
Stator water flow (normalized value)	X						< 20 %
Winding Δp (normalized value)	X						> 30 %
Individual bar flow		X[5]					Q – Q_{avg} < 10 %
Gas to water Δp (bar)			≥0.2		< 0.2		
Filter/Strainer Δp and Resin, inspection	X	X			Regular plugging / saturation		
Visual inspection stator bar		X[6]					
Hydrogen leakage (l/day)[7]	X[7]	X[7]	30 – 80		> 300	>550	
Make-up water consumption (l/week)		X	5[8]				

Δp – differential pressure(bar)
ppb – parts per billion
ppm – parts per million
T_{max}- Maximum Temperature

T_{avg} - Average Temperature
Q – Flow
Q_{avg} – Average Flow

Note:
[1] Definition of interval based on operating and Inspection experience
[2] Record of baseline under normal operating conditions
[3] Consider OEM specific limits and operating history
[4] Consider OEM specific limits
[5] See Table 3.
[6] Recommended if a chemical cleaning did not fully restore cooling in an individual bar, or where individual bar flow rates are low
[7] See Table 4
[8] Recommended value for aerated make-up water under operating conditions.

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GE Proprietary Information - Class II (Internal) US EAR - NLR

ELECTRICAL TESTING

3.2 FIELD ASSEMBLY

Part Condition: Good

Part Condition Description

A standard set of field electrical tests were performed during the outage. The filed winding resistance, AC impedance test, and 500 VDC insulation resistance and polarization index tests were performed with the field in, respectively.

The filed winding resistance marginal lower than previous measurements but compared very favorably to factory winding resistance values. The difference from test results in 2009 can be attributed to a difference between the temperature of the field and high humidity during this outage.

The AC impedance test data curve was linear, suggesting a lack of gross shorted turns while the field is in a static state. However, this test doesn't confirm whether any short turns are present in the running (dynamic) state.

The field 500 VDC insulation resistance with polarization index test result were nominal and indicated the winding insulation is in a satisfactory condition and suitable for continued service.

Background: The Generator Field Winding Resistance is measured to identify shorted turns (lower resistance), high resistance connections, broken jumpers, loose radial studs, etc. (high resistance). The Generator Field AC Impedance Test is performed on field windings to detect the presence of shorted turns or changes in the number of shorted turns in the winding. The Generator Field Insulation Resistance and Polarization Index test is used to determine the integrity of the winding insulation.

Recommendation

Perform a standard field test series at each Minor and Major outage IAW GEK 103566 Revision L. The results should be compared to previous test results for indications of winding and insulation degradation over time

MLPP_290T483_FIELD TESTING_COMPLETED.PDF

Field Measurements

Date: 03/16/18 Generator Serial No.: 290T483 Prepared by: Chau Tran
Customer: Dyengy Corp Turbine Serial No.: 270T483 FSR No.: 10301282
Manufacture: GE

DC INSULATION RESISTANCE AT 500 VOLTS

Time (min.)	R (megohms)
0.50	2730
1	10100
2	19210
3	25000
4	28300
5	33400
6	35600
7	39000
8	41000
9	44300
10	52700
PI	5.22

FIELD WINDING RESISTANCE

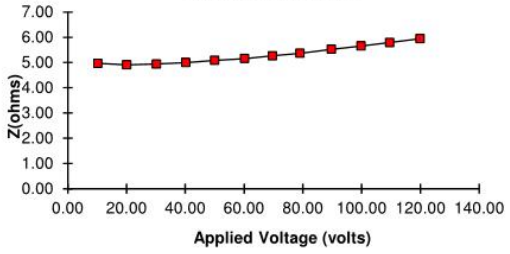
Temp (ºC)=	26.0
Resistance =	0.1395
Resistance corrected to 25 º C =	0.1390
Prior Test @25ºC	0.1428
Date	10/12/2009
Factory Resistance at 25ºC=	0.1403

FIELD IMPEDANCE CHARACTERISTICS

Applied Voltage	Amps	Z (ohms)
10.33	2.08	4.97
20.08	4.09	4.91
30.23	6.12	4.94
40.22	8.05	5.00
50.00	9.83	5.09
60.19	11.67	5.16
69.70	13.24	5.26
79.04	14.73	5.37
89.80	16.25	5.53
99.90	17.67	5.65
109.60	18.93	5.79
119.90	20.16	5.95

Temp (C)
18.0
Relative Humidity
90.0%

FIELD IMPEDANCE CHARACTERISTIC



Comments:

MEGGER ID: 1216701	DRLO ID: 1075894	Fluke ID: 121263
Cal Date : 10/18/2017	Cal Date: 1/9/2015	Cal Date: 1/9/2015
Cal Exp : 10/17/2018	Cal Exp: 1/9/2016	Cal Exp: 1/9/2015

D316303 Sheet: (a) Rev 0.0 - 05/01/05

GE Energy Proprietary Information

GENERATOR - STATOR
MAGIC INSPECTION
3.3 GENERATOR STATOR

Part Condition: Fair

CBM: NA

Part Condition Description

In beginning, the stator core lamination packages were visually inspected with the field "In-Situ" with only minor generator disassembly necessary using the high-resolution video cameras on-board the GE MAGIC robot for any indication of bent and discolored laminations as well as for spacing that would be a result of loose lamination.

However, due to discovery of loose and hollow wedges. The stator core laminate packages were visually inspected again upon the field was removed from stator.

Stator Core-iron

- Despite the minor red dusting at step-iron of both ends and various area of stator core. Also, slightly bent core-iron-laminations at a few locations the stator core-iron appeared in acceptable physical conditions.

Stator Wedges

- Minor dust was found in various body and end – wedges, which indicated there may loose and hollow wedges. Subsequently, the full wedge tightness mapping with the WITS, Wedge Inspection Testing System, confirmed where the dust came from.

Flux Probes

- Flux probe appeared to be in good shape during the MAGIC. However, the flux probe tip was found to be broken after the field was removed. This damage happened during the field removal process.

Entire stator core body was cleaned with denature alcohol before and after repaired of generator wedges. Also, the damaged flux probe was removed and replaced with new parts.

Recommendation

Conduct visual inspection of stator core by utilized MAGIC inspection method at the next minor or major outage

290T483 PHOTO LAYOUT.PDF



Image01

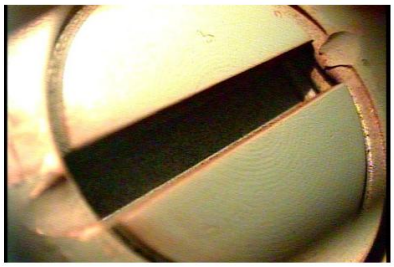


Image02



Image03



Image04

3/14/2018 DYNEGY 290T483 MOSS LANDING 1



Image05

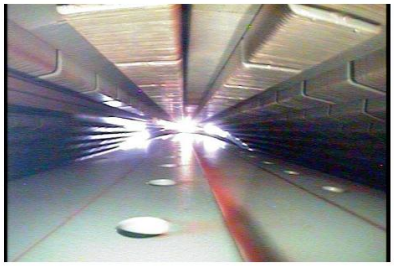


Image06



Image07



Image08



Image09



Image10

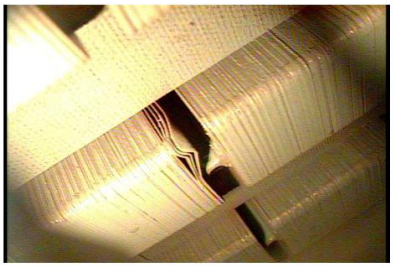


Image11

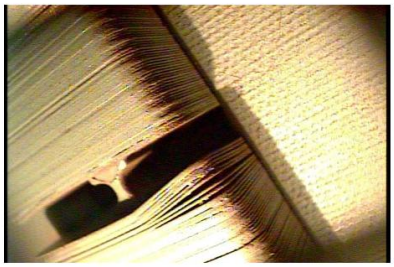


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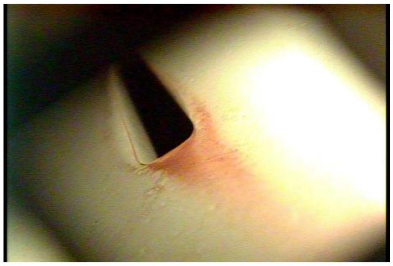


Image13



Image14



Image15



Image16



Image17



Image18



Image19



Image20



Image21

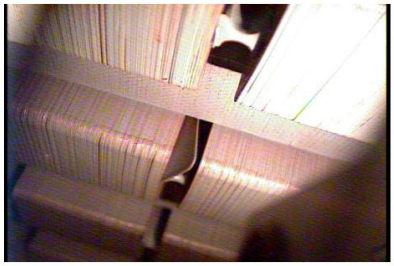


Image22



Image23

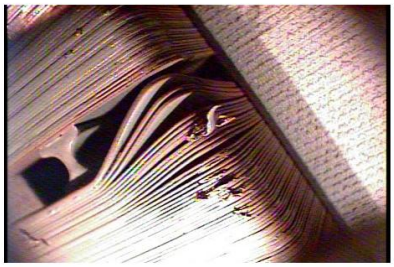


Image24



Image25



Image26



Image27

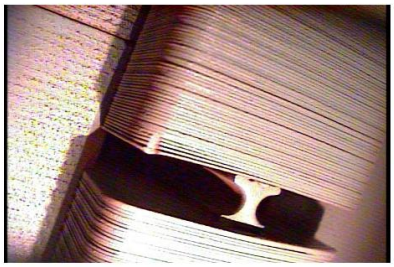


Image28



Image29



Image30



Image31



Image32



Image33



Image34



Image35



Image36



Image37



Image38



Image39



Image40



Image41



Image42



Image43

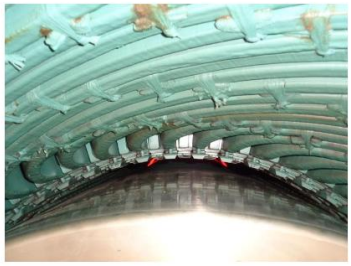


Image44



Image45



Image46



Image47



Image48



Image49



Image50



Image51



Image52



Image53



Image54

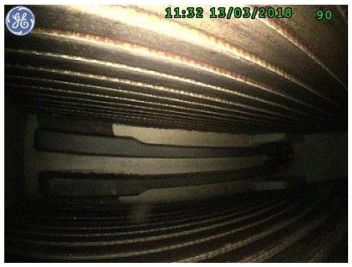


Image55



Image56



Image57



Image58



Image59



Image60



Image61



Image62



Image63



Image64



Image65



Image66



Image67



Image68



Image69

290T483 PHOTO ID.PDF

GE Energy Services

Inspection & Repair Services
General Electric International, Inc

1 River Road, Bldg. 55 Rm. 263
(518) 385-1751 , 8*235-1751
Fx: (518) 385-3178

Magic Photo I.D.Worksheet

Job # 10301282#LES-270T483-01
Generator # 290T483
Date 3-14-2018
Operator Jon Meacham
Generator Specialist Chau Tran

Location	Comments	Photo Identification	Source
Slot 30	Typical view of end wedge dowel pin with epoxy ~ CE RR	Image01	Magic Jr.
150 Deg.	Typical view of field balance weight with staking ~ 43" from CE RR	Image02	Magic Jr.
Slot 25	View of discoloration on end wedge ~ CE RR	Image03	Magic Jr.
Slot 15	Overview of step irons and endwedges ~ CE RR	Image04	Magic Jr.
234 Deg.	View of red dusting on field ~ 31" from CE RR	Image05	Magic Jr.
234 Deg.	View of red dusting on field ~ 52" from CE RR	Image06	Magic Jr.
246 Deg.	View of red dusting on field ~ 55" from CE RR	Image07	Magic Jr.
258 Deg.	View of red dusting on field ~ 50" from CE RR	Image08	Magic Jr.
306 Deg.	Typical view of CE RR	Image09	Magic Jr.
318 Deg.	View of red dusting on field ~ 56" from CE RR	Image10	Magic Jr.
Slot 57	View of bent laminations ~ 42" from CE RR	Image11	Magic Jr.
Slot 50	View of bent laminations ~ 42" from CE RR	Image12	Magic Jr.
48 Deg.	View of red dusting on field slot ~ 14" from CE RR	Image13	Magic Jr.
48 Deg.	View of red dusting on field slot ~ 37" from CE RR	Image14	Magic Jr.
Slot 44	Overview of field and stator ~ 20" from CE RR	Image15	Magic Jr.
Slot 40	Overview of stator core irons ~ 18" from CE RR	Image16	Magic Jr.
Slot 42	View of foreign material ~ 26" from CE RR	Image17	Magic Jr.
78 Deg.	View down field vent hole ~ 1" from CE RR	Image18	Magic Jr.
90 Deg.	View of red dusting of field ~ 48" from CE RR	Image19	Magic Jr.
Slot 37	Typical view of Flux Probe ~ 10" from CE RR	Image20	Magic Jr.
144 Deg.	Typical view down field vent hole ~ 164" from CE RR	Image21	Magic Jr.
Slot 22	View of bent lamination ~ 68" from CE RR	Image22	Magic Jr.
Slot 21	Typical view of end wedge dowel pin with epoxy ~ TE RR	Image23	Magic Jr.
Slot 15	View of bent laminations ~ 113" from CE RR	Image24	Magic Jr.
252 Deg.	Typical view of TE RR	Image25	Magic Jr.
252 Deg.	Overview of red dusting on field ~ 73" from CE RR looking at TE RR	Image26	Magic Jr.
Slot 4	View of foreign material in stator vent slot ~ 89" from CE RR	Image27	Magic Jr.
Slot 3	View of impact on core irons and bent laminations ~ 91" from CE RR	Image28	Magic Jr.
342 Deg.	Typical view of field balance weight with staking ~ 121" from CE RR	Image29	Magic Jr.
Slot 54	Overview of core step irons and end wedges ~ TE RR	Image30	Magic Jr.
18 Deg.	Overview of field and stator ~ 153" from CE RR looking at TE RR	Image31	Magic Jr.
Slot 38	Overview of stator core irons ~ 143" from CE RR	Image32	Magic Jr.
CE EW	Axial support hardware missing epoxy ~ 3 o'clock	Image33	Camera
CE EW	Greasing on end cap ~ 3 o'clock	Image34	Camera
CE EW	Greasing on phase connection and support ring ~ 2 o'clock	Image35	Camera
CE EW	Greasing on end cap ~ 2 o'clock	Image36	Camera

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Magic Photo I.D.Worksheet

Job # 10301282#LES-270T483-01
Generator # 290T483
Date 3-14-2018
Operator Jon Meacham
Generator Specialist Chau Tran

CE EW	Greasing on inner basket ~ 12 o'clock	Image37	Camera
CE EW	Greasing on inner basket ~ 11 o'clock	Image38	Camera
CE EW	Greasing on phase connection ~ 10 o'clock	Image39	Camera
CE EW	Typical view of core compression nut and locking tab ~ 12 o'clock	Image40	Camera
CE EW	Axial support hardware missing epoxy ~ 9 o'clock	Image41	Camera
CE EW	Greasing on phase connection ~ 8 o'clock	Image42	Camera
CE EW	Greasing on phase connection ~ 6 o'clock	Image43	Camera
CE EW	Overview of inner basket ~ 12 o'clock	Image44	Camera
CE EW	Overview of generator ~ CE	Image45	Camera
CE EW	Typical view of core compression nut and locking tab ~ 6 o'clock	Image46	Camera
CE EW	Axial support hardware missing epoxy ~ 7 o'clock	Image47	Camera
CE EW	Axial support hardware missing epoxy ~ 5 o'clock	Image48	Camera
CE RR	Main lead nut side ~ 180 Deg.	Image49	Borescope
CE RR	Main lead bolt side ~ 180 Deg.	Image50	Borescope
CE RR	Main lead bushing ~ 180 Deg.	Image51	Borescope
CE RR	Coil to coil jumper 2 to 3 ~ 180 Deg.	Image52	Borescope
CE RR	Coil to coil jumper 4 to 5 ~ 180 Deg.	Image53	Borescope
CE RR	Overview of blocking ~ 180 Deg.	Image54	Borescope
CE RR	Pole to pole jumper ~ 90 Deg.	Image55	Borescope
CE RR	View down sub slot ~ 90 Deg.	Image56	Borescope
CE RR	View down field vent hole ~ 90 Deg.	Image57	Borescope
CE RR	Stacking and turns ~ 90 Deg.	Image58	Borescope
CE RR	Overview of blocking ~ 90 Deg.	Image59	Borescope
CE RR	Main lead nut side ~ 0 Deg.	Image60	Borescope
CE RR	Coil to coil jumper 2 to 3 ~ 0 Deg.	Image61	Borescope
CE RR	Coil to coil jumper 4 to 5 ~ 0 Deg.	Image62	Borescope
CE RR	Main lead bolt side ~ 0 Deg.	Image63	Borescope
CE RR	Main lead bushing ~ 0 Deg.	Image64	Borescope
CE RR	View down sub slot ~ 270 Deg.	Image65	Borescope
CE RR	View down field vent hole ~ 270 Deg.	Image66	Borescope
CE RR	Pole to pole jumper ~ 270 Deg.	Image67	Borescope
CE RR	Stacking and turns ~ 270 Deg.	Image68	Borescope
CE RR	Overview of blocking ~ 270 Deg.	Image69	Borescope

RADIAL SLOT WEDGES

3.4 GENERATOR STATOR WEDGES

Part Condition: Fair

CBM: NA

Part Condition Description

In beginning the, a full Stator Wedge Tightness Assessment was performed on all inner body and end wedges of the stator as part of the MAGIC inspection. The wedge tightness assessment was performed by utilized the WITS. This method of wedge assessment is capable of testing end-wedge (air-gap range from 0.540 inches or larger). This WITS principle is based on applied force measurement displacement. A 600lbs of force was applied to each wedge, deflection of the wedge was being measured to determine the tightness of the wedge.

WITS Wedge Tightness Criteria:
- Loose – less than 80 (red)
- Hollow – 80 to 100 (yellow)
- Tight — 100 or greater (green)

Found 201 out of 1680 (12%) wedges were hollow and 159 out of 1680 (9%) wedges were loose.

Upon the field was removed, the wedge tightness assessment was carried out by utilized Equotip Procedure.

Equotip Wedge Tightness Criteria:
- Loose – less than or equal 549 (red)
- Hollow – from 550 to 649 (yellow)
- Tight — 650 or greater (green)

Found 372 out of 1680 (22%) wedges were hollow and 191 out of 1680 (11%) were loose.

To correct this problem the entire wedges on slot number one were removed and replaced with new inner body and end wedges and associated parts (slider, filler, and top ripple spring. Also, all the end wedges were found to be loose or hollow were removed and replace with new parts. The rest of loose and hollow inner body wedges were shimmed.

Wedge shimming is a new repair procedure that was developed by GE Engineer to repair loose or hollow inner body wedges, please see attachment for more detail of this repair procedure.

Prior of this repair, an ELCID testing of stator core was carried out and another ELCID testing was also conducted after the repair was completed

Also, the post-repair wedge tightness assessment was also conducted by both WITS and Equotip procedures.

WITS procedure found 64 of 1680 (4%) wedges were hollow and 9 out of 1680 (< 1%) wedges were loose. Equotip procedure found 100% of wedges were tight.

All replaced and wedges with shim were painted with a beige color paint for future identification.

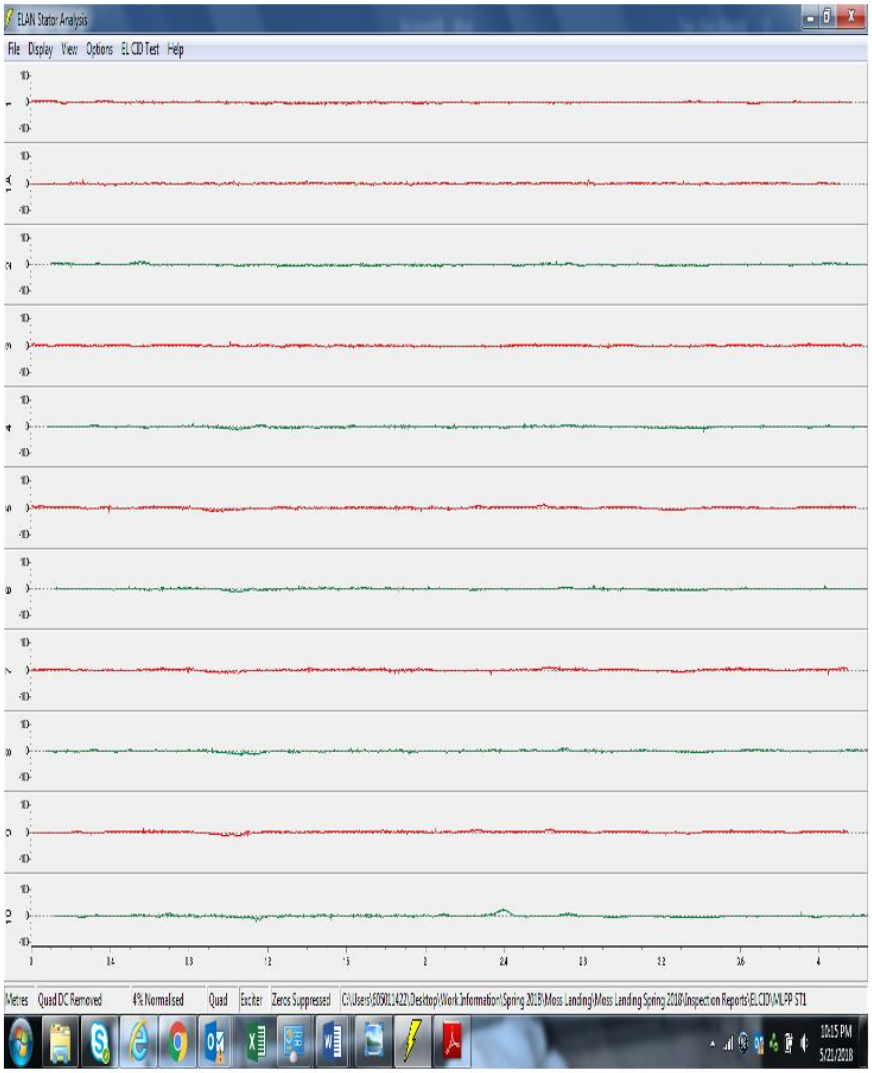
Recommendation

Perform wedge tightness assessment on a next outage opportunity

MLPP ST1 GENERATOR POST-REPAIR ELCID TEST_4-5-2018.PDF

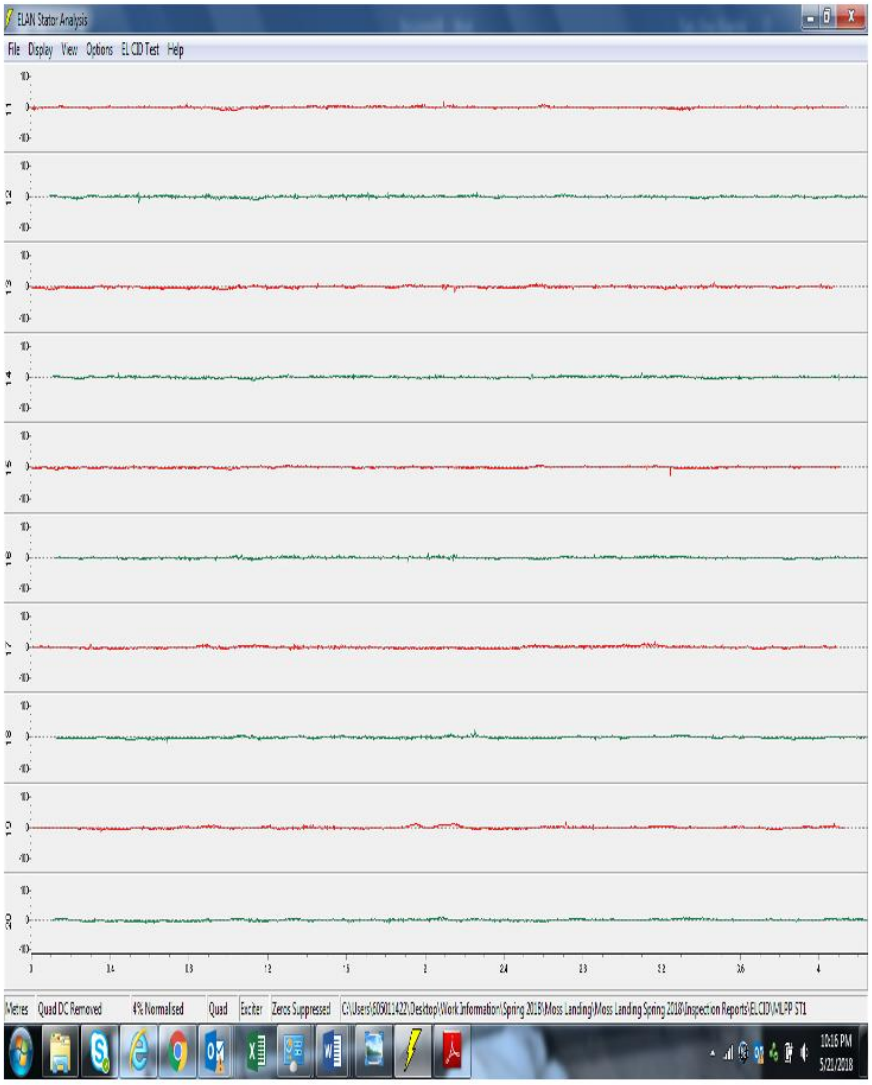
MLPP ST1 Generator Post-repair ELCID Test

Slots 1-10



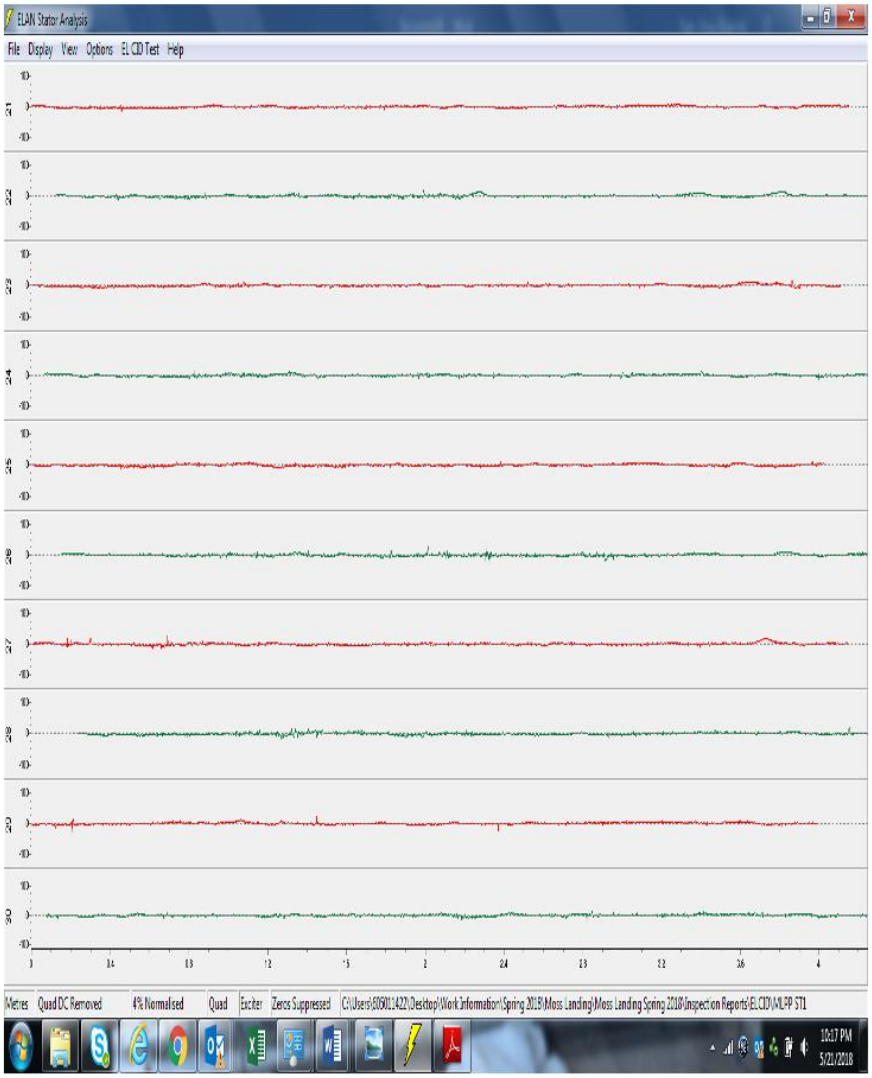
MLPP ST1 Generator Post-repair ELCID Test

Slots 11-20



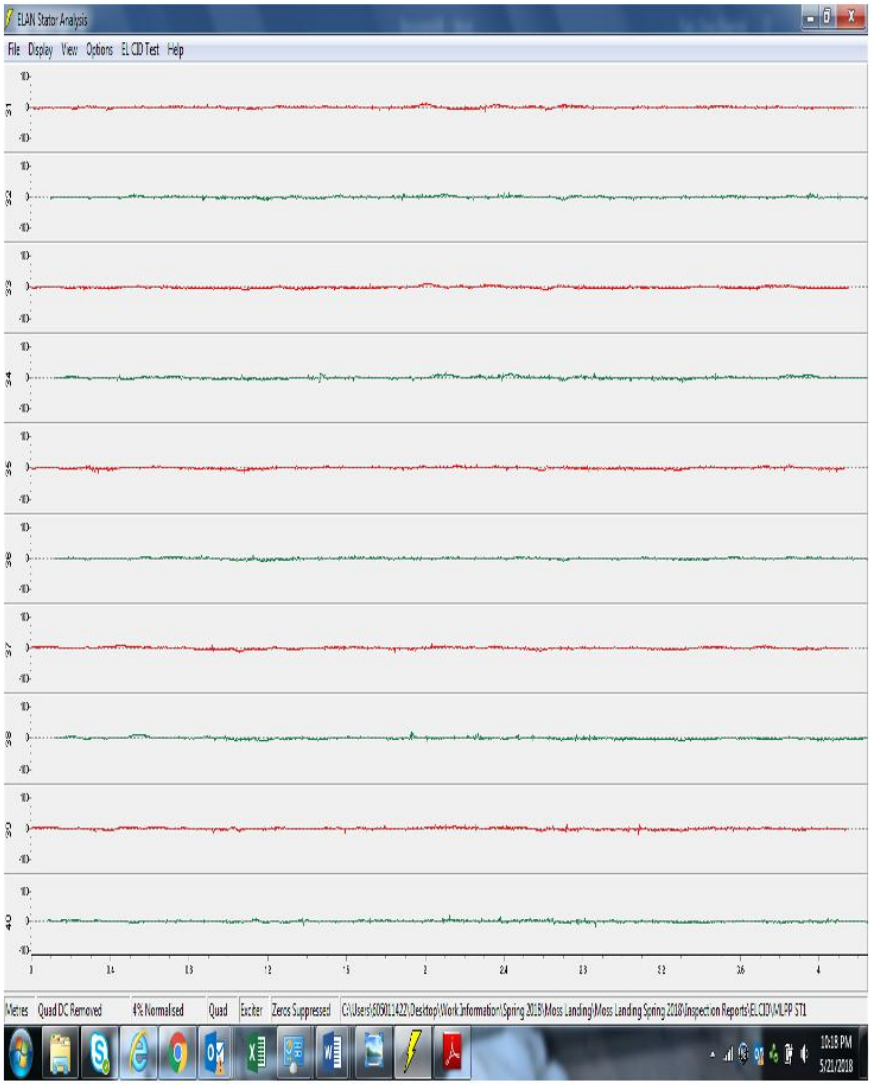
MLPP ST1 Generator Post-repair ELCID Test

Slots 21-30



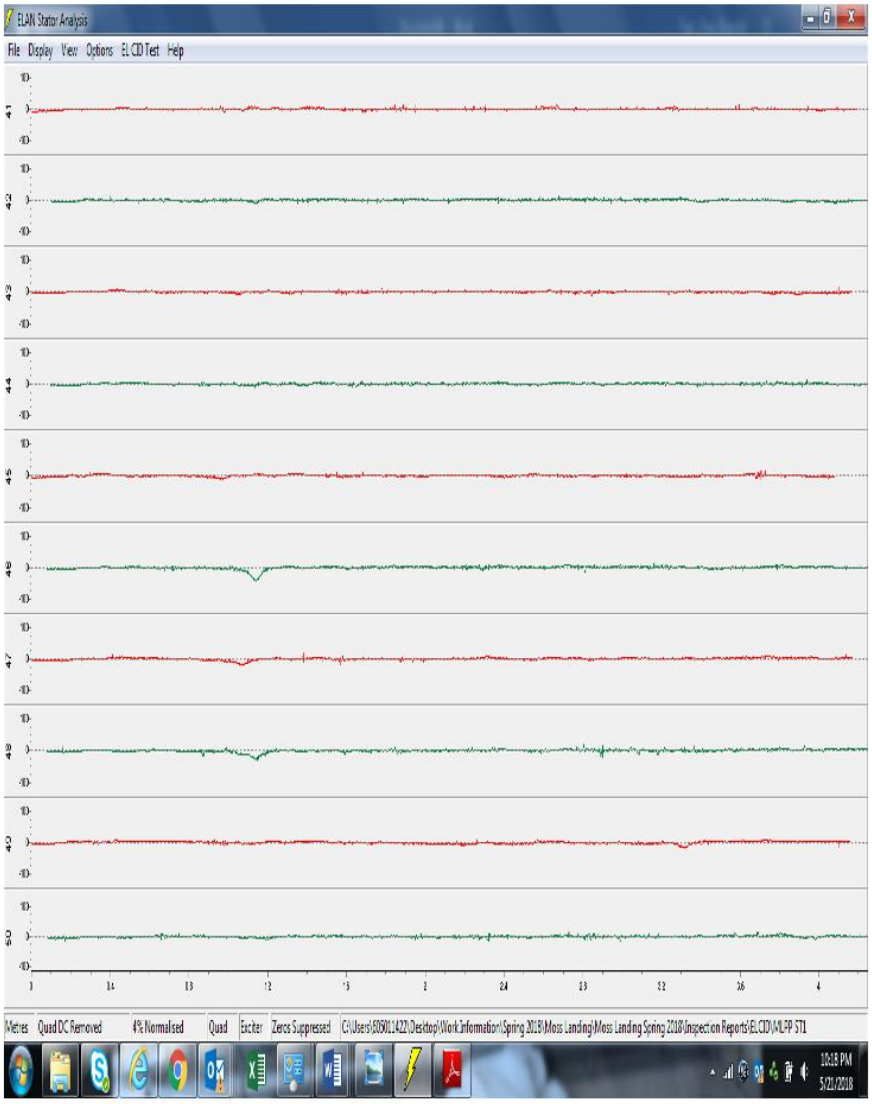
MLPP ST1 Generator Post-repair ELCID Test

Slots 31-40



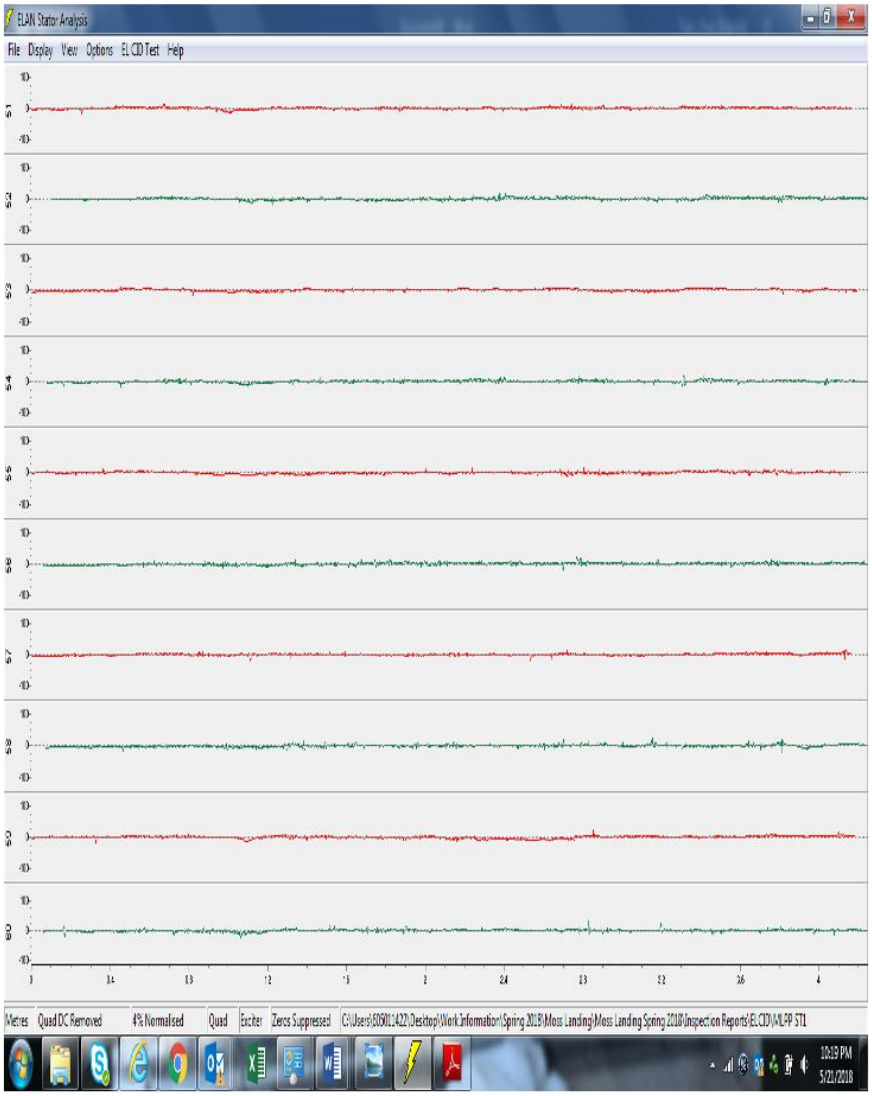
MLPP ST1 Generator Post-repair ELCID Test

Slots 41-50



MLPP ST1 Generator Post-repair ELCID Test

Slots 51-60



MLPP ST1 Generator Post-repair ELCID Test

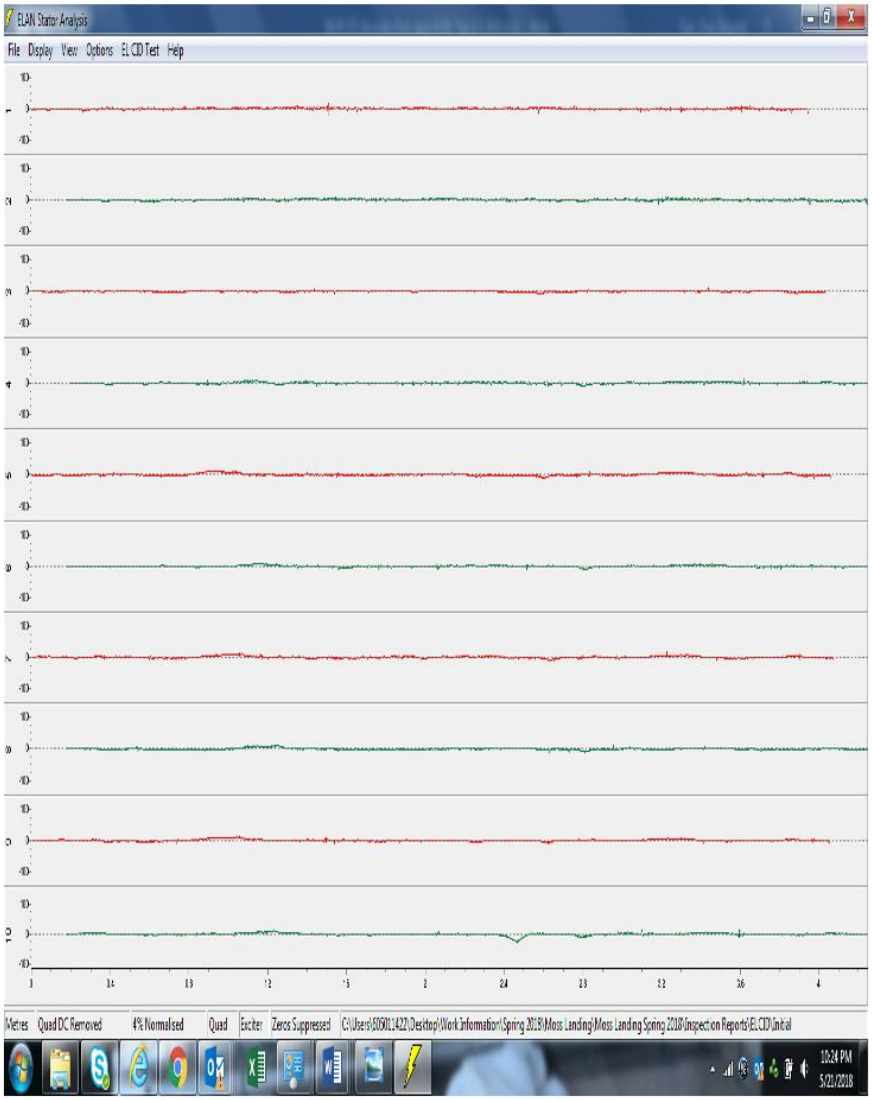
Peaked at each slot

1	1A	2	3	4	5	6	7	8	9
-19	14	-12	16	20	-15	11	-17	-13	-16
10	11	12	13	14	15	16	17	18	19
27	15	-17	-15	14	-23	13	15	24	17
20	21	22	23	24	25	26	27	28	29
-15	-14	16	12	-12	-13	-27	28	24	-26
30	31	32	33	34	35	36	37	38	39
-12	13	12	11	15	-16	9	14	18	-16
40	41	42	43	44	45	46	47	48	49
17	13	-18	14	14	15	-41	-18	-28	-16
50	51	52	53	54	55	56	57	58	59
-16	-18	-16	-17	-21	18	24	-18	23	23

MLPP ST1 GENERATOR PRE-REPAIR ELCID TEST_3-26-2018.PDF

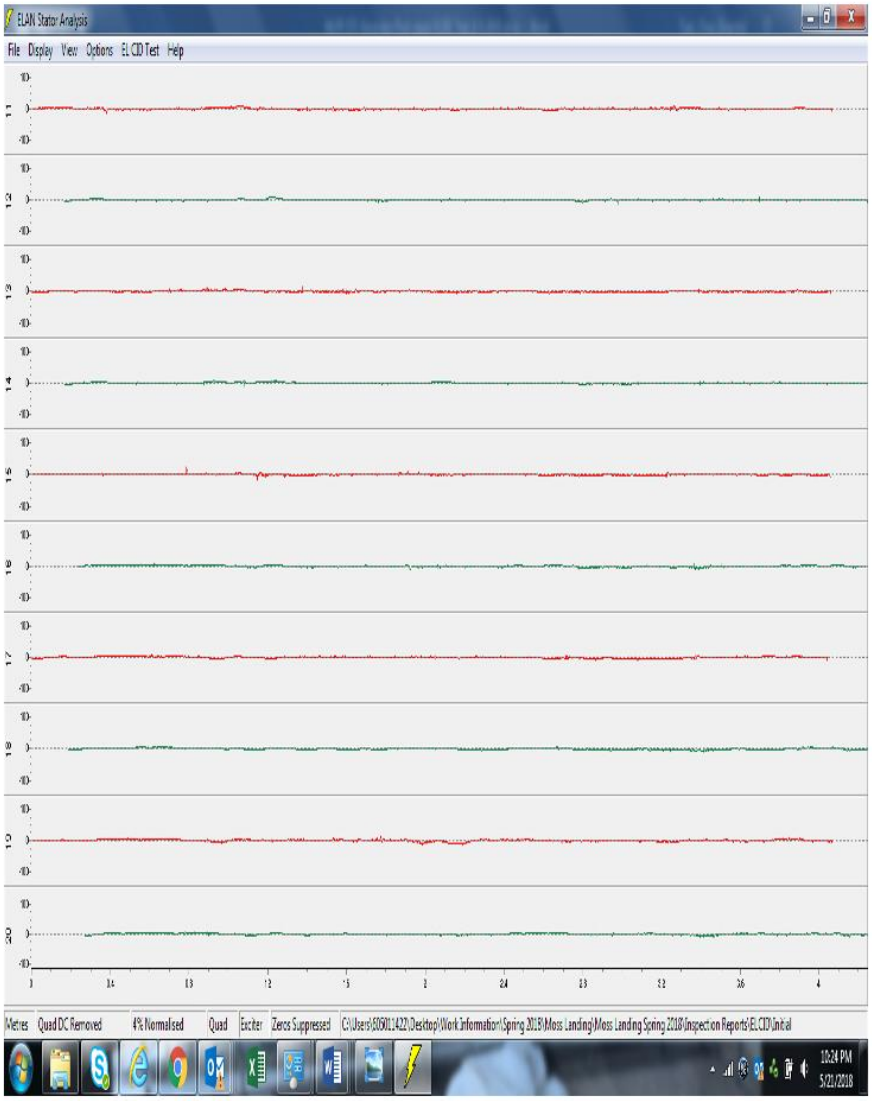
MLPP ST1 Generator Pre-repair ELCID Test

Slots 1-10



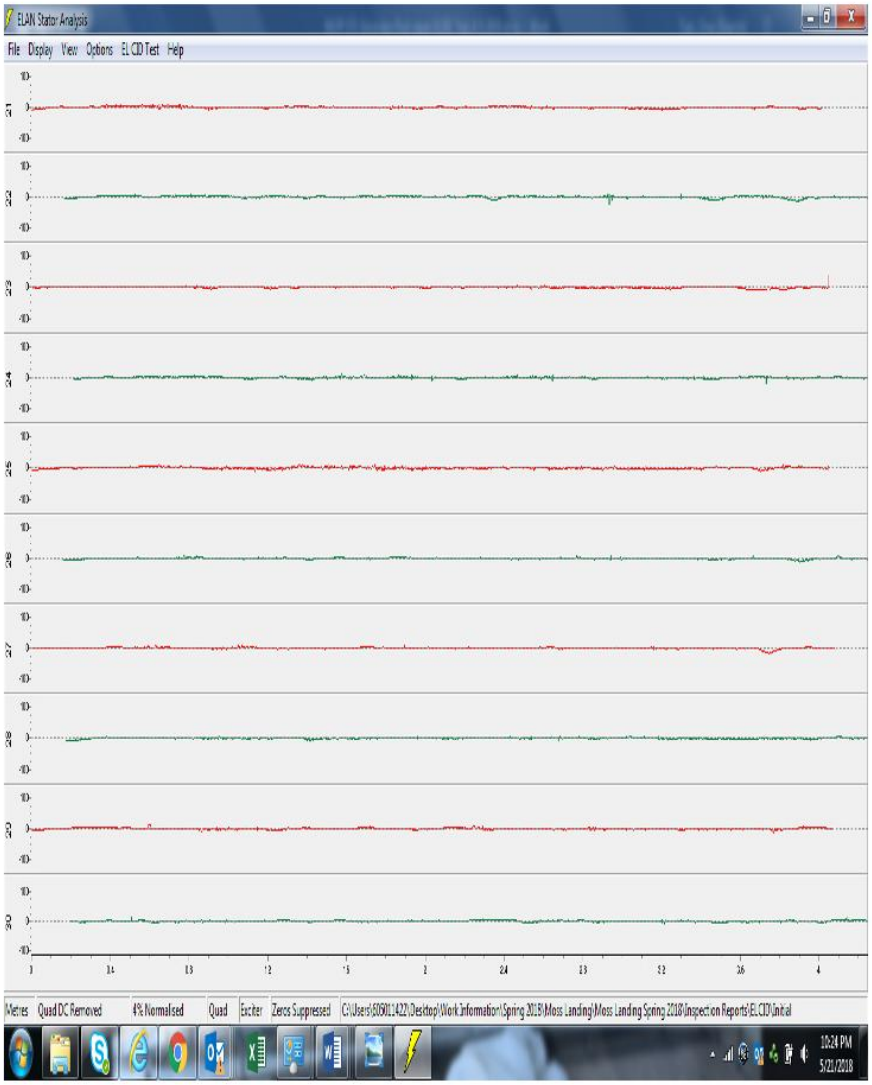
MLPP ST1 Generator Pre-repair ELCID Test

Slots 11-20



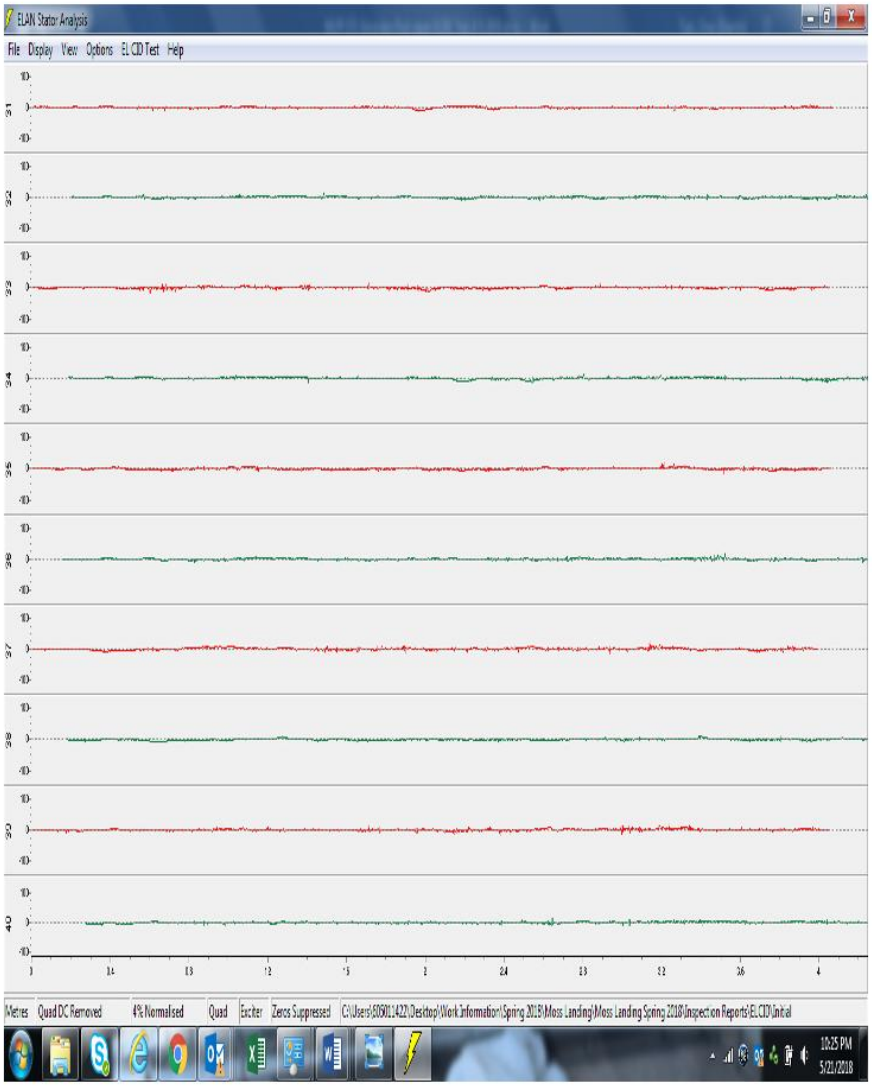
MLPP ST1 Generator Pre-repair ELCID Test

Slots 21-30



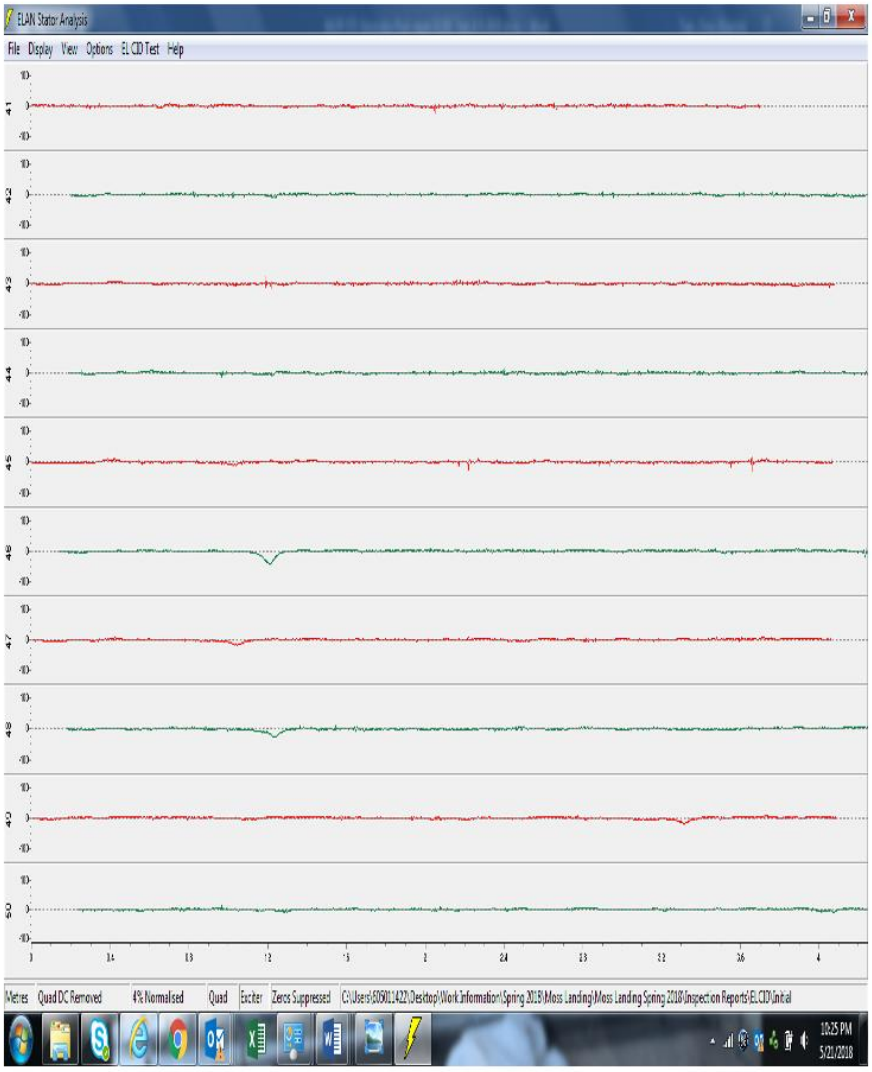
MLPP ST1 Generator Pre-repair ELCID Test

Slots 31-40



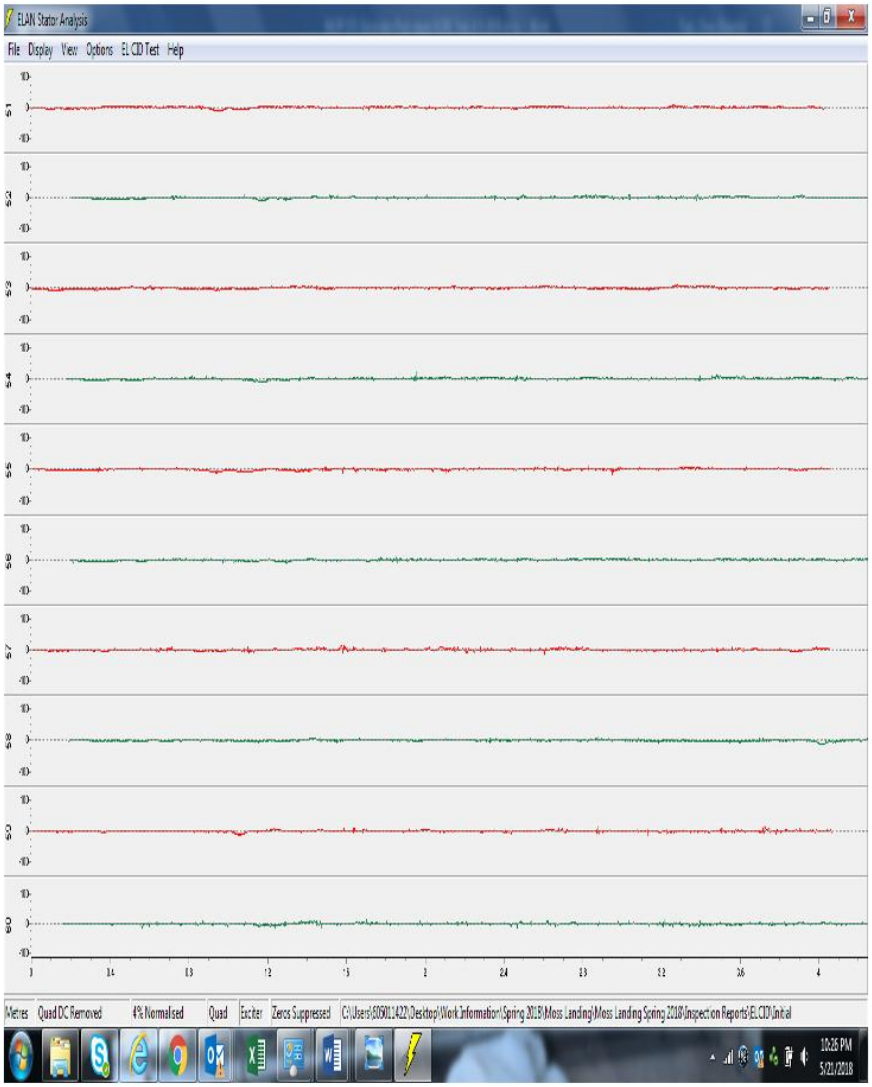
MLPP ST1 Generator Pre-repair ELCID Test

Slots 41-50



MLPP ST1 Generator Pre-repair ELCID Test

Slots 51-60



MLPP ST1 Generator Pre-repair ELCID Test

Peaked at each slot

1	2	3	4	5	6	7	8	9	10
-16	-14	9	11	13	-17	16	-11	12	-26
11	12	13	14	15	16	17	18	19	20
13	-12	14	8	23	-10	-11	-11	13	-11
21	22	23	24	25	26	27	28	29	30
-11	-22	43	-17	11	-12	-16	6	11	11
31	32	33	34	35	36	37	38	39	40
-12	14	15	13	17	16	26	10	18	14
41	42	43	44	45	46	47	48	49	50
-20	9	23	11	-27	-40	-16	-25	-17	13
51	52	53	54	55	56	57	58	59	60
-13	9	10	23	-15	10	13	-13	-14	-15

STATOR WEDGE SHIMMING FACT SHEET2 .PDF

GE Power & Water

Fact Sheet

Stator Dovetail Wedge Shimming

Loose Stator Wedges

The discovery of loose stator slot wedges on units that have been operating in the field has created the need for re-wedging the unit. Loose wedges have been attributed to allowing excessive relative movement between the core laminations, side ripple springs, and stator bars. In some cases, the magnitude and frequency of this vibratory wear has led to moderate dusting/greasing, and has also been associated with severe bar abrasion.

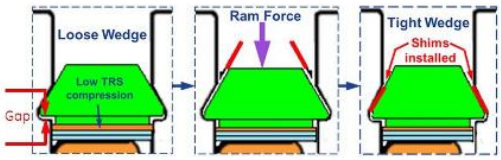
Until recently, re-wedging was the only way to correct this condition. Re-wedging is an effective method however it typically entails replacing many more wedges than necessary (to tighten wedges in the middle of the core) and time waiting for parts. In order to improve the cost and cycle of this process, GE has developed an in-situ wedge tightening process that nearly eliminates material lead times and cuts the overall cycle and cost by over 50%.

Features and Benefits

- Non-invasive, non-destructive process for tightening loose stator body wedges
- Cycle and cost reduced by 50% compared to re-wedge
- "Target Tightening" - May tighten a "pocket" of loose wedges in the center of the core without disturbing those axially outward on either end
- Number of parts required and procurement time greatly reduced
- Tighten wedges without the risk of damaging stator bars
- Re-using existing wedges eliminates break-in period associated with new wedges

Test Validation

- Extensive material testing on shimming and stator slot components
- Full scale trial completed in a 7A6 stator



Wedge Shimming Procedure

Application

- Shimming process is currently available for tightening body wedges for any generator that uses the 6" piggy back wedge system.
- There are plans to develop a method for applying shim process to end wedges with the locking dowel.
- Like current re-wedge process, must be done with rotor out.

To learn more about this product, contact your GE Power & Water sales representative or visit www.ge.com/energy

WEDGE SHIMMED AND REPLACED.PDF

Wedges ID with beige color paint for either shimmed or replaced

New flux probe installed

New end wedges replaced

Wedges were shimmed or replaced ID with beige color paint

ELECTRICAL TESTING

3.5 GENERATOR STATOR ELECTRICAL TESTING

Part Condition: Good

CBM: NA

Part Condition Description

The following standard suitability-for-service DC tests were performed on the generator stator winding and stator RTD during this outage:

- Pre-repair and Post-repair 10-minute 5000 VDC insulation resistance with polarization index (MEGGER)
- Armature winding resistance
- Pre-repair DC leakage test up to 27K Vdc
- Post-repair DC leakage test up to 36K Vdc
- Post-repair HIPOT @38K Vdc
- RTD element and resistance tests
- Pre-repair and post-repair ELCID tests

Recommendation

Conduct a standard suitability-for-service- DC test at any minor or major outage IAW GEK 103566 Rev. L

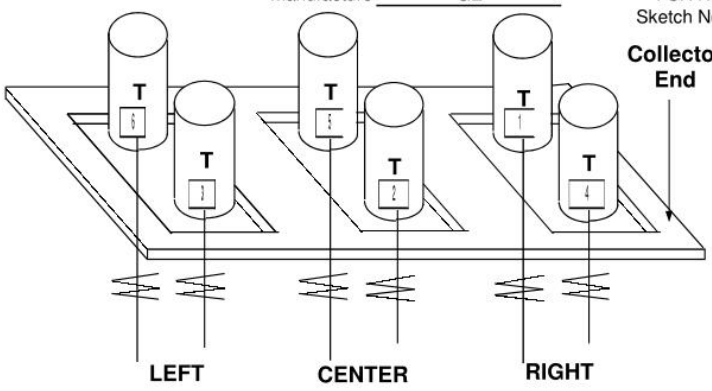
MLPP_290T483_STATOR ARMATURE WINDING RESISTANCE_COMPLETED.PDF

Armature Winding Resistance

Single Terminal

Date: 3/18/2018 | Generator Serial No.: 290T483 | Turbine Serial No. 270T483 | Manufacture GE

Prepared by: Chau Tran, Jonathan Meachan | FSR No. 10301282 | Sketch No. No



Terminal Labels	Left	Center	Right
Turbine End	6	5	1
Collector End	3	2	4

Customer Phase Designation: (A-B-C, Left-Center-Right, etc.)	T1 - T4	T2 - T5	T3 - T6
	Phase 3	Phase 2	Phase 1

Electrical Outline Drawing No.	

	T	T	T
Line Terminals	3	2	1
Neutral Terminals	6	5	4

WINDING RESISTANCE (OHMS)			LEFT T3 - T6	CENTER T2 - T5	RIGHT T4 - T1
Resistance @	32	ºC	0.001365	0.001370	0.001375
Resistance corrected to 25ºC:			0.001329	0.001334	0.001339
Factory Resistance @ 25ºC			0.001403	0.001403	0.001403
Prior test @ 25ºC, Date:		3/25/2016	0.001256	0.001256	0.001255

Comments:	
DRLO: 1216782 Cal Date: 10/17/2017 Cal Exp: 10/17/2018	

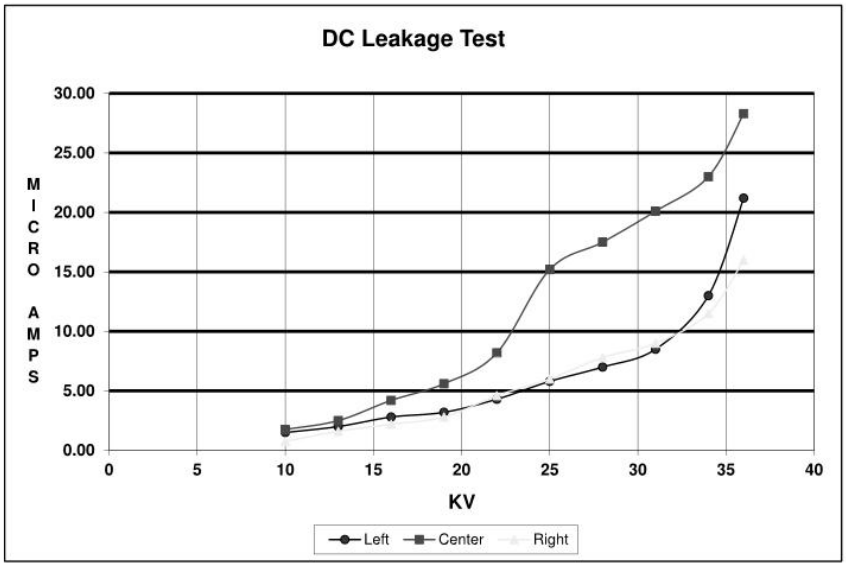
D3162041 Sheet: (a) Rev 0.0 - 05/01/05

GE Energy Proprietary Information

MLPP_290T483_STATOR DC LEAKAGE_POST REPAIR_CHART.PDF

Armature Insulation DC Leakage Test

Date(m/d/y)	4/6/208	Generator Serial No.	290T483	Prepared by	Chau Tran
Customer	Dynegy	Turbine Serial No.	270T483		Paul Tang
Manufacture	GE			FSR No.	10301282



Comments:	
Serial No. of Equipment:	1526794A
Calibration Date:	JULY 5TH 2017
Calibration Due Date:	JULY 5TH 2018

D3162044 Sheet: (b) Rev 0.0 - 05/01/05

GE Energy Proprietary Information

MLPP_290T483_STATOR DC LEAKAGE_POST REPAIR.PDF

Armature Insulation DC Leakage Test

Date(m/d/y) 4/6/208 | Generator Serial No. 290T483 | Prepared by Chau Tran
Customer Dynegy | Turbine Serial No. 270T483 | Paul Tang
Manufacture GE | FSR No. 10301282

A 10 KV DC voltage was applied to each phase with the other two phases grounded for a period of 10 minutes. Readings were recorded at specific time intervals. On the basis of these readings a time schedule is established for reading stabilized leakage currents at successively higher voltage values up to a maximum value, approximately twice line-to-line voltage. The current readings are plotted as a function of voltage to determine if a nearly linear relationship exists, indicating a constant insulation resistance as a function of voltage.

Time	Left:		Center:		Right:	
(Minutes)	µA	Megohm	µA	Megohm	µA	Megohm
0.50	6.70	1500	7.00	1400	5.20	1900
1.00	5.10	2000	5.30	1900	3.20	3100
1.50	4.00	2500	4.50	2200	2.60	3800
2.00	3.50	2900	3.75	2700	2.10	4800
3.00	3.00	3300	3.25	3100	1.80	5600
3.16	2.75	3600	3.00	3300	1.60	6300
4.00	2.50	4000	2.85	3500	1.40	7100
5.00	2.25	4400	2.75	3600	1.35	7400
6.00	2.00	5000	2.50	4000	1.30	7700
7.00	2.00	5000	2.25	4400	1.25	8000
8.00	1.75	5700	2.00	5000	1.20	8300
9.00	1.50	6700	1.75	5700	1.00	10000
10.00	1.50	6700	1.75	5700	0.75	13000
PI	3.40		3.03		4.27	
N	4		4		4	

Winding Temp (ºC)
60 f
Wet Bulb (ºF)
59
Dry Bulb (ºF)
Relative Humidity
100%

% of	Left:			Center:			Right:		
10 KV	Time	KV	µA	Time	KV	µA	Time	KV	µA
100	10'	10	1.50	10'	10	1.75	10'	10	0.75
130	13'14"	13	2.00	13'36"	13	2.50	13'36"	13	1.60
160	16'39"	16	2.80	16'39"	16	4.20	16'39"	16	2.20
190	19'21"	19	3.20	19'21"	19	5.60	19'21"	19	2.80
220	21'47"	22	4.30	21'47"	22	8.20	21'47"	22	4.60
250	24'0"	25	5.80	24'0"	25	15.20	24'0"	25	6.00
280	26'5"	28	7.00	26'5"	28	17.50	26'5"	28	7.80
310	27'59"	31	8.50	27'59"	31	20.10	27'59"	31	9.00
340	29'47"	34	13.00	29'47"	34	23.00	29'47"	34	11.50
360	31'28"	36	21.20	31'28"	36	28.30	31'28"	36	16.00
300									
320									
340									
360									
380									
	HIPOT	38	26.00	HIPOT	38	36.00	HIPOT	38	18.00

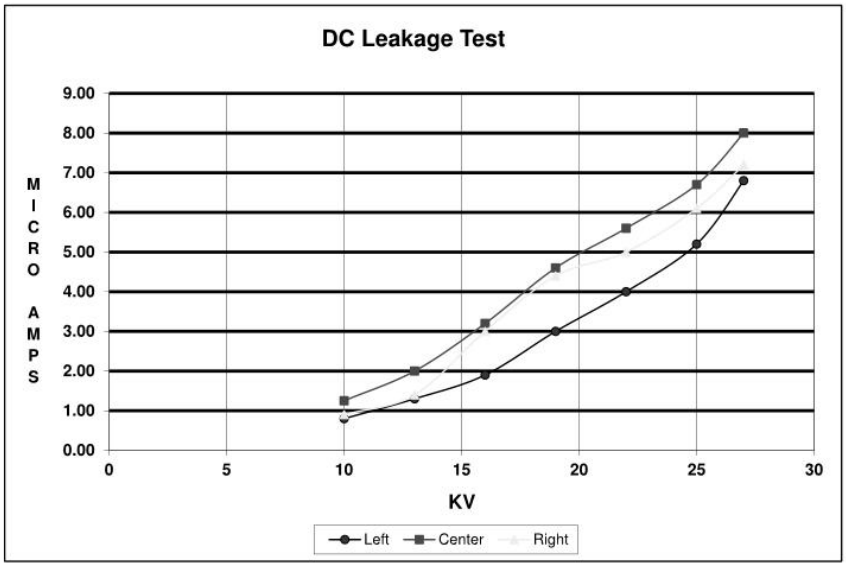
D3162044 Sheet: (a) Rev 0.0 - 05/01/05

GE Energy Proprietary Information

MLPP_290T483_STATOR DC LEAKAGE_PRE-REPAIR_CHART.PDF

Armature Insulation DC Leakage Test

Date(m/d/y) 3/18/2018 Generator Serial No. 290T483 Prepared by Chau Tran
Customer Dynegy Turbine Serial No. 270T483 Jonathan Meacham
Manufacture GE FSR No. 10301282



Comments:	
Serial No. of Equipment: Calibration Date: Calibration Due Date:	

D3162044 Sheet: (b) Rev 0.0 - 05/01/05

GE Energy Proprietary Information

MLPP_290T483_STATOR DC LEAKAGE_PRE-REPAIR.PDF

Armature Insulation DC Leakage Test
Pre-repai

Date(m/d/y) 3/18/2018 | Generator Serial No. 290T483 | Prepared by Chau Tran
Customer Dynegy | Turbine Serial No. 270T483 | Jonathan Meacham
Manufacture GE | FSR No. 10301282

A 10 KV DC voltage was applied to each phase with the other two phases grounded for a period of 10 minutes. Readings were recorded at specific time intervals. On the basis of these readings a time schedule is established for reading stabilized leakage currents at successively higher voltage values up to a maximum value, approximately twice line-to-line voltage. The current readings are plotted as a function of voltage to determine if a nearly linear relationship exists, indicating a constant insulation resistance as a function of voltage.

1

Time	Left:		Center:		Right:	
(Minutes)	µA	Megohm	µA	Megohm	µA	Megohm
0.50	4.80	2100	5.80	1700	5.00	2000
1.00	3.00	3300	3.60	2800	3.00	3300
1.50	2.30	4300	2.80	3600	2.20	4500
2.00	1.80	5600	2.60	3800	1.80	5600
3.00	1.50	6700	2.40	4200	1.60	6300
3.16	1.40	7100	2.20	4500	1.60	6300
4.00	1.20	8300	2.20	4500	1.40	7100
5.00	1.20	8300	1.80	5600	1.20	8300
6.00	1.10	9100	1.60	6300	1.10	9100
7.00	1.00	10000	1.60	6300	1.00	10000
8.00	0.90	11000	1.50	6700	1.00	10000
9.00	0.80	13000	1.50	6700	0.90	11000
10.00	0.80	13000	1.25	8000	0.90	11000
PI	3.75		2.88		3.33	
N	8		3		4	

Winding Temp (ºC)
26

Wet Bulb (ºF)
59

Dry Bulb (ºF)
65%

Relative Humidity
80%

% of	Left:			Center:			Right:		
10 KV	Time	KV	µA	Time	KV	µA	Time	KV	µA
100	10'	10	0.80	10'	10	1.25	10'	10	0.90
130	13'14"	13	1.30	13'27"	13	2.00	13'36"	13	1.40
160	17'22"	16	1.90	16'21"	16	3.20	16'39"	16	3.00
190	20'25"	19	3.00	18'55"	19	4.60	19'21"	19	4.40
220	23'12"	22	4.00	21'12"	22	5.60	21'47"	22	5.00
250	25'46"	25	5.20	23'18"	25	6.70	24'0"	25	6.10
270	28'9"	27	6.80	25'14"	27	8.00	26'5"	27	7.20
	HIPOT			HIPOT			HIPOT		

D3162044 Sheet: (a) Rev 0.0 - 05/01/05

GE Energy Proprietary Information

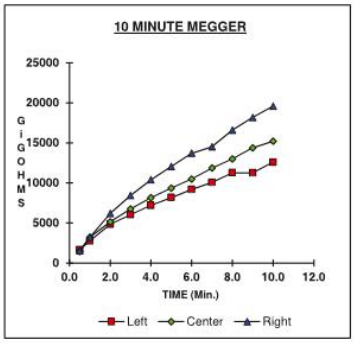
MLPP_290T483_STATOR ARMATURE INSULATION RESISTANCE_POST REPAIR.PDF

Armature Insulation Resistance / HV Proof Tests
Post Repair

Date 4/6/2018 | Generator Serial No. 290T483 | Prepared by Chau Tran
Manufacture GE | Turbine Serial No. 270T483 | Paul tang
Customer Dynegy ML | FSR No. 10301282
Sketch No. No

DC INSULATION RESISTANCE (Megohms)
Voltage: 5000 Volts

Time(Min)	Left	Center	Right
0.5	1650	1440	1520
1.0	2760	3210	3220
2.0	4830	5140	6190
3.0	6030	6750	8430
4.0	7180	8140	10380
5.0	8170	9350	12040
6.0	9170	10480	13700
7.0	10060	11850	14510
8.0	11270	13000	16590
9.0	11250	14380	18170
10.0	12590	15200	19580
PI	4.6	4.7	6.1



101T3628

HIGH VOLTAGE PROOF TEST (One minute test on each phase with the other phases grounded.)
Rated Voltage (E): 18 KV

Test At:
60 Hz
0.1 Hz
DC

RECOMMENDED TEST VOLTAGES
AC Test Voltage (60 Hz): 27.0 KV
AC Test Voltage (0.1 Hz): 43.9 KV
DC Test Voltage: 45.9 KV DC
Actual Test Voltage: 28 KV DC

	LEFT	CENTER	RIGHT
Passed test at			
Failed test at			

Winding Temp: 26 C
Wet Bulb: 65 F
Dry Bulb: 59 F
Rel. Humidity: 100 %

AC HIPOT DATA (60 HZ) if applicable:
Test Set:

	LEFT	CENTER	RIGHT
Excitation Voltage			
Current (Amps)			

Polarization Index
PI = 10min/1min

Comments:
Serial No. of MEGGER: 1216781
Calibration Date: 10/18/2017
Calibration Due Date: 10/17/2018

D3162043 Sheet: (a) Rev 0.0 - 05/01/05

GE Energy Proprietary Information

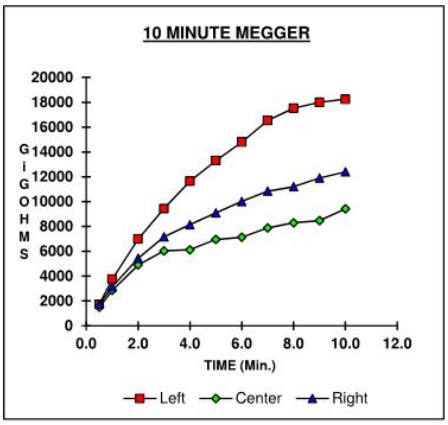
MLPP_290T483_STATOR ARMATURE INSULATION RESISTANCE_PRE-REPAIR.PDF

Armature Insulation Resistance / HV Proof Tests
Pre-repair

Date 3/18/2018 | Generator Serial No. 290T483 | Prepared by Chau Tran
Manufacture GE | Turbine Serial No. 270T483 | Jonathan Meacham
Customer Dynegy ML | FSR No. 10301282
Sketch No. No

DC INSULATION RESISTANCE (Megohms)
Voltage: 5000 Volts

Time(Min)	Left	Center	Right
0.5	1730	1500	1750
1.0	3760	2850	3140
2.0	6990	4900	5440
3.0	9440	6030	7160
4.0	11640	6120	8140
5.0	13300	6950	9090
6.0	14800	7130	10010
7.0	16530	7880	10840
8.0	17510	8300	11200
9.0	18000	8470	11890
10.0	18250	9420	12400
PI	**4.9**	**3.3**	**3.9**



HIGH VOLTAGE PROOF TEST (One minute test on each phase with the other phases grounded.)
Rated Voltage (E): 18 KV

Test At:
60 Hz
0.1 Hz
DC

RECOMMENDED TEST VOLTAGES
AC Test Voltage (60 Hz): 27.0 KV
AC Test Voltage (0.1 Hz): 43.9 KV
DC Test Voltage: 45.9 KV DC
Actual Test Voltage: 28 KV DC

	LEFT	CENTER	RIGHT
Passed test at			
Failed test at			

Winding Temp: 27 C
Wet Bulb: 65 F
Dry Bulb: 59 F
Rel. Humidity: 90 %

AC HIPOT DATA (60 HZ) if applicable:
Test Set:

	LEFT	CENTER	RIGHT
Excitation Voltage			
Current (Amps)			

Polarization Index
PI = 10min/1min

Comments:
Serial No. of MEGGER: 1216781
Calibration Date: 10/18/2017
Calibration Due Date: 10/17/2018

D3162043 Sheet: (a) Rev 0.0 - 05/01/05

GE Energy Proprietary Information

END WINDING

3.6 GENERATOR END-WINDING

Part Condition: Fair

CBM: NA

Part Condition Description

During the visual inspection of generator end-winding connection ring, minor greasing was found on various CE ties.

Also, minor turbine lubrication oil contamination was found at the bottom of the generator collector

end. Connection ring movement and lubrication oil ingress probably was caused of this greasing condition.

All minor greasing ties were cleaned with denature alcohol and then inspected for any damage. No damage was found on any ties. Red eye epoxy was applied to all ties with grease.

Connection ring area was also thoroughly cleaned with denature alcohol.

TIL 1398-2R1 Inspection of stator end-winding axial support system hardware. The requirement of TIL was not 100% completed, only upper half was completed on previous outage. All support hardware on lower-half were covered with epoxy, Astro 6000, during this outage.

Recommendation

Conduct visual inspection of CE at the next opportunity outage.

CE TIES.PDF



As-found Ce ties with grease



As-found CE tie with grease



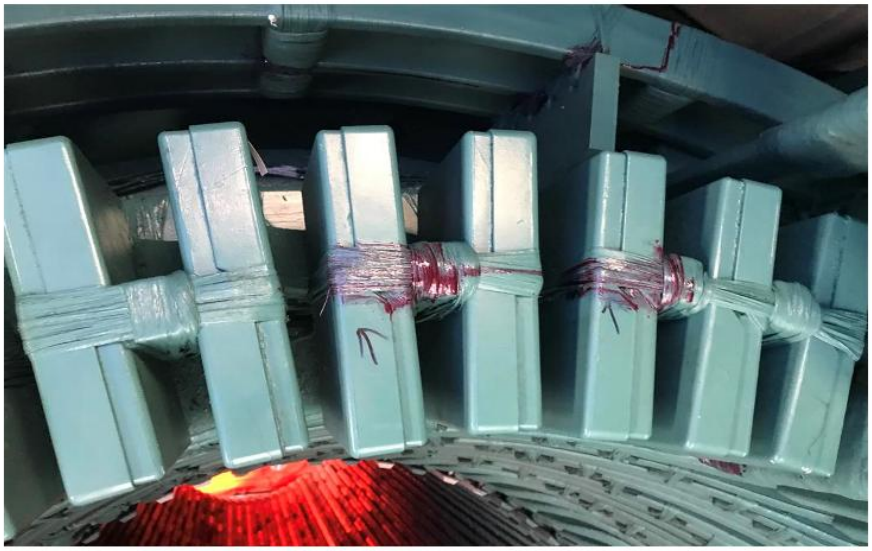
As-found CE tie with grease



As-found CE tie with grease



Red-eye epoxy applied on CE ties



Red-ey epoxy applied on CE ties

HV BUSHINGS

3.7 HIGH VOLTAGE BUSHING

Part Condition: Fair

CBM: NA

Part Condition Description

High voltage bushings (HVB's) were inspected internally from inside of the high voltage bushing box. No visible signs of cracks or chips were observed on any of the stand-off insulators or any of the high voltage bushings. The wells that surrounded the high voltage bushings were filled with Viscasil. This verified that there was no hydrogen leakage from the flanges of the high voltage bushings. The condition of the Viscasil was clean, no sign of oil contamination found.

Recommendation

High voltage bushings and stand-off insulators should be inspected for cleanliness as well as the presence of any cracks or deformities at any minor or major outage.

HVB.PDF



As-found condition of HVB Viscasil



As-found Viscasil in HVB well appeared to be clean